Who Could Have Rational Expectations?

Bounds Across People and Information Sets

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Abstract

Tests of rational expectations ask whether the hypothesis can be rejected. We ask how much of a population could satisfy it. The weight on a component satisfying the rational-expectations restriction, mixed with an unrestricted alternative, is bounded above, in closed form, but never below. In the Health and Retirement Study, we reject a necessary implication of rational expectations and bound the population share compatible with it. At most 96.71 percent can be rationalized given reported survival beliefs, and 91.72 percent once beliefs must also be efficient with respect to own sex, age, health and smoking. Testing whether that share is below one is exact in finite samples under within-cell independence, and measuring how far below is not. The bound moves with the information asserted, the outcome forecast, the population and the follow-up observed. Death-index records and later interviews raise the finest-rung non-rationalizable share from 6.50 to 8.28 percent, still rising.

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1 Introduction

Empirical work on expectations usually asks whether rational expectations can be rejected, and the answer is usually yes. Household forecast errors are correlated with characteristics the respondent knew when making the forecast (Souleles, 2004). Individual professional forecasters overreact to news, while consensus forecasts underreact (Coibion and Gorodnichenko, 2015; Bordalo et al., 2020). Elicited survival probabilities depart from realized mortality in ways that differ with smoking and sex (Hurd, 2009), and rational expectations about earnings is rejected for workers without a college degree and not for college graduates (D'Haultfœuille et al., 2021).

The data behind such tests are a stated probability that an event will happen and, when follow-up is long enough, whether it happened. Let p_i be a person's subjective probability of a binary event, Y_i its realization and I_i the information the person has when forecasting. Rational expectations in the sense of Muth (1961) is the restriction $p_i = \mathbb{E}[Y_i \mid I_i]$. The analyst does not observe I_i, but can name characteristics Z_i the person surely knows, such as their age or whether they smoke. Because the person also knows their own belief, rational expectations implies $\mathbb{E}[Y_i \mid p_i, Z_i] = p_i$ however much private information I_i contains. Among people who hold the same belief and share the same Z, the event must occur at that rate. We use only this calibration restriction, observe beliefs as rounded reports, and allow for outcomes that are not yet resolved. Satisfying the restriction does not show that a person formed rational expectations, and we make no assumption about preferences or about how people come to know what they know.

A rejection of the restriction rules out rational expectations for the population as a whole, under the maintained measurement and information assumptions. It does not say how widespread the incompatibility is. We ask how large a share of a population could satisfy the restriction when the rest is left entirely unrestricted.

The list of characteristics is the analyst's assumption. It requires only that the listed characteristics be part of what the person knows, so a shorter list is a weaker assumption. Only characteristics fixed before the report qualify, and knowing a characteristic does not mean using it. In the application the list starts empty and adds in turn the respondent's own sex, age group, self-rated health and smoking. These are alternative assumptions about the same moment, and they do not describe learning over time.

We characterize the largest share of the joint distribution of beliefs and outcomes that can satisfy the restriction. Its identified set runs from zero to an upper bound. The lower end is zero in every dataset, because an unrestricted alternative can always mimic rational behaviour, so only the upper bound carries information. We call it the maximum rationalizable share. Its complement is the minimum share of the population that no rational-expectations representation can accommodate. The analysis concerns the observable implications of rational expectations, and it identifies neither which individuals violate them nor what the violation costs.

The bound is computed cell by cell. Within a cell defined by the belief and the asserted characteristics, the component satisfying the rational-expectations restriction must have mean equal to the belief and the remainder may have any mean. That restriction caps the rational mass in closed

form, and the population bound averages the caps. The cap equals one only where the realized rate equals the belief, and it falls linearly on either side, steeply near zero and one. Reported probabilities are coarse, so beliefs enter as respondent-specific intervals. Conditioning cells on the report requires that the report carry no information about the outcome beyond what the respondent knows, and without that assumption the bound is lower. Unresolved outcomes are carried as mass allocated in the way most favourable to rationality. Because the cap is concave in the realized rate, asserting that agents know more weakly lowers the bound, so bounds across nested information sets are ordered in the population and not only in the estimator.

Testing whether the bound equals one is much easier than estimating how far below one it lies. The rational-expectations restriction fails for the whole population as soon as one cell's realized rate is incompatible with every belief the cell admits, and each cell contributes two one-sided hypotheses about a binomial proportion. Under independent sampling the test is exact in finite samples and needs neither a linear program nor a bootstrap. Bounding the magnitude aggregates evidence across every cell and weight at once, and the resulting functional is directionally but not fully differentiable, a class in which the standard bootstrap is consistent only where the functional is fully differentiable (Fang and Santos, 2019). This asymmetry helps explain why the literature reports rejections of rational expectations far more often than the share of a population they implicate.

We apply the framework to survival beliefs in the Health and Retirement Study, where respondents report the percent chance of reaching an individually tailored target age and the outcome is later observed. The calibration restriction is rejected for the population as a whole, also under two treatments that relax the within-cell independence the binomial test assumes. Respondents whose reports admit no more than a three to five percent chance are already known to have reached their target age about eighteen percent of the time. Respondents whose reports admit nothing below ninety-five percent are known to have reached it between thirty-two and thirty-eight percent of the time, and counting every unresolved case as a survivor still leaves them short of the lowest belief their reports admit.

The magnitude is positive and small. Conditioning on the reported belief alone, at most 96.71 percent of the weighted population reporting on a grid finer than the focal one could be rationalized, so at least 2.79 percent could not, at ninety-five percent confidence. Treating respondents whose reports admit no point belief in the way most favourable to rationality extends the statement to everyone eligible, with a floor of 2.32 percent. Requiring beliefs to be efficient with respect to the respondent's own sex, age, self-rated health and smoking lowers the maximum rationalizable share to 91.72 percent and raises the floor to 6.45 percent, from under three people in a hundred to more than six. The rise does not come from cutting the data more finely. The respondent's own sex barely moves the bound, and a population that is rational by construction, evaluated on the same cells, produces about a quarter of the rise. Each rung detects a discrepancy between beliefs and outcomes that coarser cells conceal, and not the predictive power of the variable added.

The share also depends on the event forecast and on the population. On the same respondents, beliefs about working full time after sixty-two leave 11.38 percent non-rationalizable where survival

beliefs leave 0.45, and the survival bound varies by an order of magnitude across the subsamples that different questions define.

It depends as well on how much of the forecast future the analyst has seen. Holding the 2004 cohort, its beliefs and its weights fixed and recoding the outcome with the evidence dated by each year, the base-rung share rises from 0.13 percent in 2010 to 2.50 in 2022 and 3.29 with all current sources, and the finest-rung share from 0.99 to 8.28. The exact test already rejects with evidence dated by 2008, when more than nine outcomes in ten are unresolved, because a few cells at the tails fail as soon as their outcomes start to resolve. The incompatible share accumulates as outcomes arrive. Part of the rise comes from deaths recorded by the National Death Index and survivals confirmed after 2022, which the standard RAND file does not carry. With that file alone the two shares are 2.24 and 6.50 percent. Information assumed for the agent and information observed by the analyst are two ordered dimensions of the same identification problem.

The estimand differs from that of the closest work. Tests of rationalizability give a population-level verdict on whether beliefs and realizations can be reconciled at all (D'Haultfœuille et al., 2021). Switching models estimate time-varying shares of agents who use each of a few fully specified forecasting rules, from survey responses (Branch, 2004) or from aggregate inflation series (Cornea-Madeira et al., 2019), and their forward-looking rule is distinct from rational expectations. In choice data, Kitamura and Stoye (2018) test whether a population of utility maximizers can generate observed choice frequencies, and Caliari and Petri (2024) bound from below the fraction of decision makers who must be rational. The closest econometric object is a sharp bound on a population share obtained by optimizing over every latent structure the data admit, as in the bound of Torgovitsky (2019a) on the share of unemployment persistence due to state dependence, and the complement of our share is a mixture index of fit in the sense of Rudas et al. (1994). We supply a bound of this kind for expectations data. It is sharp in the population and leaves the non-rational component unrestricted, and the monotonicity result gives the sequence of bounds across information sets its content. The framework extends to several outcomes at once and characterizes when requiring rationalizability in every domain restricts more than the domain-by-domain bounds.

Section 2 reviews related work, Section 3 develops identification, Section 4 sets out the two inference problems, and Sections 5 and 6 present the data and the results.

2 Related Literature

Tests of rational expectations on survey forecasts ask whether the hypothesis can be rejected. Panel tests on individual professional forecasts that correct for aggregation bias and correlated errors do not reject the rationality of price forecasts (Keane and Runkle, 1990). Coibion and Gorodnichenko (2015) regress consensus forecast errors on revisions to measure information rigidity, and Bordalo et al. (2020) find overreaction for individual forecasters alongside underreaction in the consensus. In household data, forecast errors are correlated with characteristics the respondent knows when forecasting (Souleles, 2004). Apparent irrationality can sometimes be rationalized by asymmetric

loss (Elliott et al., 2005), a defence that is weaker for probability reports about binary personal events than for point forecasts. None of this work reports what fraction of the population a rejection implicates.

The measurement tradition elicits probabilistic expectations in household surveys (Manski, 2004), and the resulting probabilities predict outcomes where respondents hold private information, as with survival and retirement (Hurd, 2009). In the Health and Retirement Study, average survival probabilities to seventy-five are close to life-table rates, those to eighty-five exceed them, and individual reports predict mortality after controlling for self-rated health (Hurd and McGarry, 2002).

The econometric ancestor of the share is contaminated sampling, in which a known bound on the probability of contamination yields bounds on the uncontaminated distribution (Horowitz and Manski, 1995). Here the logic is inverted, with the restriction fixed and the mixing weight bounded. That inversion is the mixture index of fit of Rudas et al. (1994), the smallest share of a population that must be set aside for a model of a contingency table to fit exactly, with the remainder unrestricted. The minimum non-rationalizable share is an index of that kind for the restriction rational expectations places on probabilistic beliefs, extended to beliefs observed as intervals and to censored outcomes, with an exact test and inference for the bound. Without further restrictions the components and weights of a finite mixture are not identified. An exclusion restriction yields a sharp identified set, and point identification of a two-component mixture needs conditions such as dominance of one component in each tail (Henry et al., 2014). Three or more conditionally independent measurements per unit identify the components nonparametrically (Bonhomme et al., 2016), and grouped fixed-effects estimators classify units consistently as the number of periods grows (Bonhomme and Manresa, 2015; Bonhomme et al., 2022). Here each respondent contributes one belief and one realization, and the alternative population is unrestricted and can reproduce any rational cell exactly. The share therefore has only an upper bound and individual membership is not identified.

The bound belongs to the literature that computes sharp identified sets by optimizing over every latent structure consistent with the data. Mogstad et al. (2018) bound policy-relevant treatment parameters by searching over the marginal treatment response functions that reproduce a set of instrumental-variable estimands. Torgovitsky (2019b) characterizes sharp identified sets by asking whether some distribution of unobservables can generate the observed distribution, and Torgovitsky (2019a) uses linear programming over latent potential outcome sequences to bound the share of unemployment persistence due to state dependence. The maximum rationalizable share is an object of the same kind, and its program separates across cells with a closed-form solution, which is what permits an exact finite-sample test. Restrictions that link cells, such as monotonicity of the alternative population in the reported probability, would remove the closed form and call for the linear-programming formulations of these papers.

What agents know has a longer history in work that separates heterogeneity agents can forecast from uncertainty they cannot (Cunha et al., 2005). There a candidate information set is valid

when the part of realized outcomes it leaves unexplained does not predict choices. The condition used here has the same structure with the elicited probability in place of the choice, since relative to the asserted information the realization minus the report must have mean zero. Cunha et al. (2005) also find that person-specific information sets are generally not identified and only their distribution is, which parallels the result that a population share can be bounded while individual membership cannot be recovered.

Two literatures come closest on expectations themselves. The first characterizes when a distribution of beliefs and a distribution of realizations can be rationalized at all. Rationalizability is equivalent to the realizations being a mean-preserving spread of the beliefs, and it can be tested with marginal distributions from unmatched datasets (D'Haultfœuille et al., 2021). That test gives a population-level verdict, and for a binary outcome it compares the mean belief with the mean realization. When beliefs and realizations are matched person by person, as here, a stronger restriction is available and it supports a share. An earlier partial-identification treatment bounds features of the realized distribution under a best-case rationality hypothesis with categorical predictions (Das et al., 1999), without the mixture object. The second literature estimates time-varying shares of agents who choose among a few fully specified forecasting rules. Branch (2004) estimates such shares from individual responses to the Michigan survey, and Cornea-Madeira et al. (2019) estimate the share of forward-looking fundamentalists from aggregate United States inflation. Both distinguish their forward-looking rule from rational expectations, and their shares depend on the forms assumed for every rule.

Reported probabilities are coarse, and a reported zero taken literally imposes an absolute restriction. The measurement model follows Manski and Molinari (2010) in inferring a respondent's rounding from their pattern of answers across questions and converting point reports into intervals. Probed after their first answer, nearly half of Health and Retirement Study respondents to a module on late-onset dementia reported their probability as a range, which Giustinelli et al. (2022) read as an imprecise belief. Once Dutch survival expectations are treated as partially identified, women's expectations can no longer be rejected as consistent with life tables (Bissonnette and de Bresser, 2018).

Decision-making quality has been studied as an individual characteristic for choice consistency (Choi et al., 2014) and for forecasting accuracy in tournaments (Mellers et al., 2015), and elicited behavioural biases are moderately stable within person (Stango and Zinman, 2020) and correlated with one another (Stango and Zinman, 2023). The analogue for expectations would need repeated belief and realization pairs for the same person in several domains, which the available panels do not supply.

Finally, the magnitude is a functional of estimated cell means and weights that is Hadamard directionally but not fully differentiable, the setting in which Fang and Santos (2019) show that the standard bootstrap is consistent only at points of full differentiability. Related procedures test whether a linear system with estimated components admits a solution, with the right-hand side estimated (Andrews et al., 2023; Fang et al., 2023) or with the coefficients estimated as well (Goff

and Mbakop, 2025; Bai et al., 2026). These tests address existence. Here existence is tested exactly in finite samples under independence within cells, and magnitude is bounded with the numerical delta method of Hong and Li (2018), implemented with multiplier draws.

3 Framework and Identification

3.1 Setup

Data. For each individual i the survey records a reported probability r_i that a binary event will occur within a stated horizon, and $Y_i \in \{0, 1\}$ indicates whether it does. In the application r_i is the percent chance, stated in 2004, of living to an individually assigned target age, and $Y_i = 1$ if the individual reaches it. The survey also records a vector W_i of characteristics, here the respondent's sex, age group, self-rated health and current smoking, which takes finitely many values. Two features of the data separate what is recorded from what the theory restricts. First, Y_i is not always resolved by the end of follow-up. Each individual is a known success ($R_{1i} = 1$), a known failure ($R_{0i} = 1$) or unresolved ($U_i = 1$), with $R_{1i} + R_{0i} + U_i = 1$, and 30.5 percent of the survival outcomes in the estimation sample are unresolved. Second, r_i is recorded on a coarse grid and need not equal the belief. The reporting rule of Section 5 maps each report into an interval $[a_i, b_i]$ that contains the belief.

The belief. Let $p_i \in [0, 1]$ denote the individual's subjective probability that $Y_i = 1$. Because the event is binary, this number is the individual's whole subjective distribution over the outcome, a Bernoulli whose mean fixes its distribution function, so a median, a mode or a full distribution would add nothing. The data reveal p_i only up to the interval $[a_i, b_i]$.

Where rationality enters. Let I_i denote the information available to the individual when the report is made, a σ-field that the analyst does not observe, and nothing below depends on how it came about. For a binary event, rational expectations in the sense of Muth (1961) is the restriction

$$p_i \;=\; \mathbb{E}[\,Y_i \mid I_i\,] \;=\; \mathbb{P}(Y_i = 1 \mid I_i),$$

where $\mathbb{P}$ is the objective probability of the event given what the individual knows. This is the only economic restriction used. Muth's hypothesis is broader, since it also ties expectations to the predictions of the relevant economic theory.

The asserted information set. Let Z_i be a subvector of W_i that the analyst asserts the individual knew at the time of the report. An individual knows their own belief, so the maintained assumption is $\sigma(p_i, Z_i) \subseteq I_i$. The inclusion may be strict, and I_i may contain much that the analyst never observes. By the law of iterated expectations, rational expectations with respect to I_i then implies

$$\mathbb{E}[\,Y_i \mid p_i, Z_i\,] \;=\; p_i,$$

so private information cannot reconcile a violation of this coarser condition. With Z empty it says $\mathbb{E}[Y \mid p] = p$, so among individuals who hold the same belief the event occurs at that rate. Each variable added to Z imposes the same restriction within finer groups, so asserting more can only lower the bound (Theorem 1).

An admissible Z must be fixed before the report and known to the individual when they make it, which excludes variables recorded later and ones the individual may not know, such as an undiagnosed condition. The variables the individual used to form the belief are unobserved, and knowing a variable does not imply using it. A calibration failure within cells of Z means that the belief did not use Z correctly, while calibration relative to Z says nothing about the rest of I_i. The condition is necessary for rational expectations and never sufficient.

The application uses five nested choices. The first, Z_0, is empty, and the others add in turn the respondent's own sex, age group, self-rated health and smoking. Write $\mathcal{Z}$ for the finite set of values of Z and $X_i = (p_i, Z_i)$ for the conditioning cell.

Reporting information. The empirical cells group respondents by their report as well as by Z, and conditioning on the report needs its own assumption. Let T_i denote reporting information, the report r_i and, where it is used to define cells, the reporting grid g_i that Section 5 infers from the respondent's other answers. The empirical model extends the mixture (3) of the exact benchmark below to the vector (Y, p, Z, T),

$$P^T_{\text{obs}} \;=\; \pi\, P^T_{\text{RE}} + (1-\pi)\, P^T_A, \qquad \mathbb{E}_{P^T_{\text{RE}}}[\, Y \mid p, Z, T\,] = p, \qquad P^T_A \text{ unrestricted}, \tag{1}$$

where P^T_{RE} is the law of the component satisfying the rational-expectations restriction. Because p is recorded only up to $[a_i, b_i]$ and some outcomes are unresolved, P^T_{obs} is a complete-data law that the data restrict only through the reports, the intervals and the resolved outcomes, and the bounds maximize over every such law consistent with the record. The restriction on P^T_{RE} follows from rational expectations when, for individuals in that component, reporting information does not predict the outcome beyond what they know, $\mathbb{E}[Y_i \mid I_i, T_i] = \mathbb{E}[Y_i \mid I_i]$, since then $\mathbb{E}[Y_i \mid p_i, Z_i, T_i] = p_i$ by the law of iterated expectations. This holds in particular when T_i is a function of the respondent's own answers in the same interview and those answers belong to I_i. It would fail if the way a respondent reports revealed health information they do not have, such as undiagnosed cognitive decline that affects both how they round and how long they live. Nothing is assumed about P^T_A. The reported estimates condition on the belief group, a function of r_i alone, and use g_i only to bound the belief, so they require (1) with $T_i = r_i$. The sharp partition of Appendix F conditions on g_i as well and requires it with $T_i = (r_i, g_i)$. Without (1), respondents in different report groups who share a latent belief fall in the same cell, discrepancies across report groups can offset one another, and the bound becomes a linear program over where each latent belief lies. Section 6 reports the bound under that weaker assumption.

The exact benchmark. Propositions 1 to 3 treat the benchmark in which the belief is recorded exactly, $r_i = p_i$, and every outcome is resolved. Beliefs then lie in a finite set $\mathcal{P} \subset [0,1]$, since reports are recorded on a grid, so (Y, p, Z) lives on the finite sample space $\{0,1\} \times \mathcal{P} \times \mathcal{Z}$. Let P_{obs} denote the population joint distribution of (Y, p, Z) on that space, which the data reveal directly in this benchmark. Every other distribution P below is a probability distribution on the same space, and each conditional expectation is taken under the law named in its subscript. Cells with zero probability play no role, so conditional expectations given X are cell means. Propositions 4 and 5 drop both simplifications. The data then restrict the joint law of (Y, p, Z) only through the intervals $[a_i, b_i]$ and the resolved outcomes, and the bounds maximize over every complete-data law consistent with what is recorded.

Definition 1 (Calibration relative to an asserted information set)**.** Let $\mathcal{R}(Z)$ denote the set of distributions P on $\{0,1\} \times \mathcal{P} \times \mathcal{Z}$ such that

$$\mathbb{E}_P[Y \mid p, Z] = p \qquad P\text{-almost surely.} \tag{2}$$

A distribution P is calibrated relative to Z if $P \in \mathcal{R}(Z)$.

Condition (2) says that among individuals who hold the same belief and share the same value of Z, the event occurs at that rate. It is a necessary implication of rational expectations when the individual knows (p, Z), by the law of iterated expectations. Conversely, a distribution satisfying it admits an observationally equivalent rational-expectations representation with information set $\sigma(p, Z)$. The converse concerns the existence of a representation, and not the information individuals actually held or how they produced their forecasts. A distribution is *rationalizable* relative to Z when it admits this representation.

Calibration is distinct from accuracy. Accuracy measures how close a forecast comes to the realization under some loss function, and it rewards knowing more. Calibration asks only whether the event occurs at the stated rate among individuals who state that rate, and a respondent who reports the unconditional base rate is calibrated while knowing nothing. Two respondents can report the same probability and experience different outcomes. Their realized forecast errors differ, but a single realization does not tell us whether either formed a rational-expectations belief. Calibration restricts conditional outcome frequencies in the population and is not a person-level diagnosis. Calibration on the belief alone is a weak restriction, and enlarging Z strengthens it.

Suppose the observed distribution can be written as a mixture of a component that satisfies Definition 1 and a component the analyst is unwilling to restrict:

$$P_{\text{obs}} \;=\; \pi\, P_{\text{RE}} \;+\; (1-\pi)\, P_A, \qquad P_{\text{RE}} \in \mathcal{R}(Z), \qquad P_A \text{ unrestricted.} \tag{3}$$

Here P_{RE} is a distribution of (Y, p, Z) that satisfies the restriction and P_A is any distribution on $\{0,1\} \times \mathcal{P} \times \mathcal{Z}$. The decomposition is an existence statement about an unobserved partition of the population. Any feasible decomposition can be given a latent-type form. Draw a type $\tau \in \{R, A\}$

with $\mathbb{P}(\tau = R) = \pi$ before the event, then draw (p, Z) and Y from P_{RE} for type R and from P_A for type A. An individual of type R whose information is $\sigma(p, Z, \tau)$ then holds the belief $p = \mathbb{P}(Y = 1 \mid p, Z, \tau = R)$, so the belief equals the objective probability given that individual's information, as rational expectations requires. The type-A component may itself look calibrated.

Proposition 1 (Identified set)**.** *Under* (3) *the identified set for* π *is the interval* $[\,0,\ \bar{\pi}(Z)\,]$*, where* $\bar{\pi}(Z) = \sup\{\pi : (3)$ *admits a solution*$\}$*. The lower bound equals zero for every* P_{obs}*.*

The lower bound is uninformative because P_A is unrestricted and may itself satisfy (2), so a world in which nobody belongs to the calibrated component and everybody merely looks calibrated is never refuted. The estimand $\bar{\pi}$ is the *maximum rationalizable share* and $1 - \bar{\pi}$ is the *minimum non-rationalizable share*, the smallest mass of the population that every decomposition must leave outside the component satisfying the restriction. Both are measured in units of population mass, and neither identifies which individuals belong to either component.

3.2 Identification

Fix a cell $x = (p, z)$ with $P_{\mathrm{obs}}(X = x) > 0$ and let $m(x) = P_{\mathrm{obs}}(Y = 1 \mid X = x)$ be the realized rate within it.

Proposition 2 (Sharp cell bound)**.** *The largest mass share within cell* x *that can satisfy* (2) *is*

$$\bar{\lambda}(p, m) \;=\; \min\left\{1,\ \frac{m}{p},\ \frac{1-m}{1-p}\right\}, \tag{4}$$

with a ratio whose denominator is zero read as $+\infty$*, and this value is attained.*

Within the cell the rational component must have mean p, while the other component may have any mean in $[0, 1]$, so $m = \lambda p + (1 - \lambda)a$ for some $a \in [0, 1]$. Maximizing λ subject to $a \in [0, 1]$ delivers (4). The function is a tent in m that equals one when $m = p$, rises with slope $1/p$ below the peak and falls with slope $-1/(1 - p)$ above it (Figure 1). It is concave in m, which drives the monotonicity result. A report near zero restricts most steeply when the event occurs even less often than reported, and a report near one when it occurs even more often.

Proposition 3 (Population bound)**.** *Let* F_X *denote the distribution of cells under* P_{obs}*, and for a cell* $X = (p, z)$ *write* $p(X) = p$ *for its reported belief. Then*

$$\bar{\pi}(Z) \;=\; \mathbb{E}_{F_X}\left[\,\bar{\lambda}\big(p(X),\, m(X)\big)\,\right], \tag{5}$$

and the bound is sharp, in the sense that a pair (P_{RE}, P_A) *attains it.*

Sharpness follows because the constraints in (3) separate across cells. The rational component's conditional law given the cell is restricted only through its mean, so the cell-level maxima can be attained simultaneously, and the rational component's marginal over cells is then determined by the attained weights. The proof is in Appendix A.

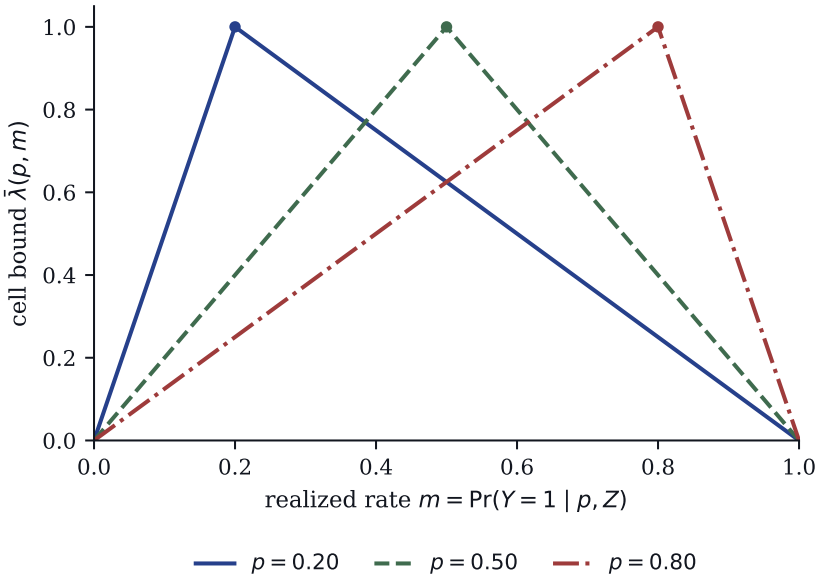


Figure 1: The sharp cell bound as a function of the realized rate

Notes. Each curve plots $\bar{\lambda}(p, m)$ from (4) against the realized rate m for one reported belief p, given in the legend. The bound equals one only where the realized rate coincides with the reported belief, shown by a dot. The left arm, where the event occurs less often than reported, is steeper for small p, and the right arm is steeper for p close to one.

The bound counts people, while a calibration error measures probability. Writing $d(x) = |m(x) - p(x)|$ for the calibration discrepancy in a cell, (4) gives $1 - \bar{\lambda} = d/p$ when $m < p$ and $1 - \bar{\lambda} = d/(1-p)$ when $m > p$, so the minimum non-rationalizable share rescales the discrepancy by the belief on the side where the shortfall occurs. It is never below the average discrepancy $\mathbb{E}_{F_X}[d(X)]$, and no fixed multiple of that average bounds it. If every individual reports a probability $\varepsilon > 0$ for an event that never occurs, the average discrepancy is ε, while no calibrated component of positive mass exists and the minimum non-rationalizable share is one. A small average error is compatible with a population no part of which satisfies the calibration restriction. In the application, with intervals and censoring treated the same way in both, the average discrepancy is 2.96 percentage points at the base rung and 5.76 at the finest, against minimum non-rationalizable shares of 3.29 and 8.28 percent.

Two features of survey data change the cell bound. Probabilities are reported on a coarse grid, so the belief is known only to lie in a respondent-specific interval $[a_i, b_i]$ inferred from the respondent's own answering pattern, and the bound maximizes over the beliefs that interval admits.

Proposition 4 (Interval beliefs)**.** *Let a cell consist of individuals with the same reporting information and the same value of Z, and suppose every belief in it is known only to lie in a common*

interval $[a, b]$. *Under* (1) *the sharp cell bound is*

$$\bar{\lambda}(a, b, m) = \begin{cases} 1, & m \in [a, b], \\ m/a, & m < a, \\ (1-m)/(1-b), & m > b. \end{cases} \tag{6}$$

The tent of Proposition 2 becomes a plateau. On the plateau the bound is locally constant in m, so the finite-sample bias of the plug-in estimator at a kink largely disappears. Interval beliefs make the estimator better behaved and the bound less informative.

Some realizations are unobserved. A respondent who is still alive and has not yet reached the target age has an undetermined survival outcome. Within a cell, let m_1 denote the share of respondents known to have $Y = 1$ and u the share whose outcome is unresolved, so that the share $1 - m_1 - u$ is known to have $Y = 0$. Treating unresolved mass as freely allocable enlarges the set of realizations compatible with rationality.

Proposition 5 (Censoring)**.** *For a cell as in Proposition 4, with belief interval* $[a, b]$*, known success mass* m_1 *and unresolved mass* u*,*

$$\bar{\lambda} = \begin{cases} 1, & [a, b] \cap [m_1,\, m_1 + u] \neq \emptyset, \\ (m_1 + u)/a, & a > m_1 + u, \\ (1 - m_1)/(1 - b), & b < m_1. \end{cases} \tag{7}$$

Theorem 1 (Information-set monotonicity)**.** *If* Z' *contains every variable in* Z*, written* $Z \subseteq Z'$*, then* $\bar{\pi}(Z') \leq \bar{\pi}(Z)$*.*

The argument is Jensen's inequality applied to a concave function. The bound $\bar{\lambda}(p, \cdot)$ is a minimum of two affine functions of m and therefore concave, and refining the conditioning set replaces $m(p, Z)$ by a collection of cell means whose average given (p, Z) is $m(p, Z)$. Hence $\mathbb{E}[\bar{\lambda}(p, m(p, Z')) \mid p, Z] \leq \bar{\lambda}(p, m(p, Z))$, and integrating gives the result. With interval beliefs and censoring the bound is concave in the cell's outcome shares and can only fall when the belief interval shrinks, and a refined cell's interval lies inside the interval of the cell it refines, so the same conclusion holds (Appendix A). The ordering $\bar{\pi}(Z_0) \geq \bar{\pi}(Z_1) \geq \cdots$ holds in the population, so an observed non-monotonicity is an estimation problem.

Proposition 6 (Follow-up monotonicity)**.** *Fix the population, the event, the cells, the belief intervals and the weights, and let additional follow-up resolve outcomes that were unresolved without changing any outcome already known. Then the bound computed from the later data is no larger than the bound computed from the earlier data, in every cell and in total.*

Resolving an outcome narrows the cell's feasible outcome range $[m_1, m_1 + u]$, and (7) is the largest value of the bound over that range, so it cannot rise. The sharp version holds for the same

reason, since the set of complete-data laws consistent with the data contracts. Theorem 1 orders bounds by what individuals are asserted to know, and Proposition 6 orders them by how much of the forecast future the analyst has observed. Both move the minimum non-rationalizable share in the same direction, so a share computed before every outcome resolves is a lower bound on what longer follow-up of the same people would show. Neither ordering applies to confidence bounds, which can move either way.

The hypothesis that the entire population could be calibrated restricts each cell separately. Index the cells with positive probability by $c = 1, \ldots, C$, and write $[a_c, b_c]$ for the belief interval of cell c and $m_1(c)$ and $u(c)$ for its known-success and unresolved shares.

Proposition 7 (Feasibility of universal calibration)**.** $\bar{\pi}(Z) = 1$ *if and only if, for every cell* c*,*

$$m_1(c) \;\leq\; b_c \qquad \textit{and} \qquad m_1(c) + u(c) \;\geq\; a_c. \tag{8}$$

Feasibility here means compatibility of the whole population with the necessary calibration implication, under the maintained information and reporting assumptions. Rejecting it rules out rational expectations for the population as a whole under those assumptions, while passing it would not establish rational expectations. The first inequality says the cell cannot produce more successes than the most permissive admissible belief allows. The second says that even counting every unresolved case as a success, the cell cannot produce too few. Proposition 7 turns a statement about a population aggregate into a finite collection of statements about single binomial proportions, which allows an exact test.

4 Testing Existence and Estimating Magnitude

Whether $\bar{\pi} < 1$ is an exact finite-sample question when observations within a cell are independent, and it reduces to a finite collection of one-sided binomial hypotheses. How far below one $\bar{\pi}$ lies is a partial-identification problem whose inference must cover every cell and every cell weight at once.

4.1 Existence

By Proposition 7, the rational-expectations restriction can hold for the whole population if and only if every cell satisfies (8), so rejecting it requires only that *one* inequality fail. For each respondent let R_1 indicate that the outcome is known to have occurred, R_0 that it is known not to have occurred and $U = 1 - R_1 - R_0$ that it is unresolved, so that $P_{\text{obs}}(R_1 = 1 \mid c) = m_1(c)$ and $P_{\text{obs}}(U = 1 \mid c) = u(c)$. The two conditions in cell c are

$$P_{\text{obs}}(R_1 = 1 \mid c) \leq b_c \qquad \text{and} \qquad P_{\text{obs}}(R_0 = 1 \mid c) \leq 1 - a_c, \tag{9}$$

the second being equivalent to $P_{\text{obs}}(R_1 = 1 \mid c) + P_{\text{obs}}(U = 1 \mid c) \geq a_c$.

Each of (9) is a one-sided hypothesis about a single proportion, conditional on the cell count, with exact p-values from the binomial distribution,

$$p_c^{\text{high}} = \mathbb{P}[\text{Bin}(n_c, b_c) \geq k_{1c}]\,, \qquad p_c^{\text{low}} = \mathbb{P}[\text{Bin}(n_c, a_c) \leq k_{1c} + k_{uc}]\,. \tag{10}$$

Here n_c is the number of respondents in cell c, k_{1c} the number known to have reached the outcome and k_{uc} the number whose outcome is unresolved. The null is the intersection of all $2C$ inequalities, so the Holm step-down procedure controls the family-wise error rate in finite samples under arbitrary dependence across the tests. The test needs neither resampling nor estimated cell weights, and the multiplicity burden is mild because a single decisive cell suffices.

Cells with fewer than twenty-five respondents are not tested, since a handful of observations cannot refute the inequality at any useful level. Restricting the family to a subset of cells leaves the global test valid and can only cost power. Exactness requires independent Bernoulli observations *within* a cell, and household surveys interview more than one member of a household, so the test is also reported under two treatments that allow for that dependence.

4.2 Magnitude

When the cells partition the population, $w_c = P_{\text{obs}}(c)$ is the population share in cell c and $\bar{\lambda}_c$ is the sharp cell bound of Section 3 evaluated at cell c,

$$1 - \bar{\pi} \;=\; \sum_c w_c \left(1 - \bar{\lambda}_c\right), \tag{11}$$

with equality. It becomes an inequality only when inference replaces each term by a lower bound or restricts attention to a subset of cells.

One route to a one-sided lower bound on (11) inverts the same exact binomials cell by cell and combines them with exact bounds on the weights. It is valid in finite samples and very conservative, because the multiplicity correction runs over every cell proportion and weight and each term is bounded at its own worst case, as if all cells could be extreme at once. In the application it yields a lower bound of 0.40 percent against an unweighted point value of 6.39 percent at the base rung.

Design-based replication applied to the aggregate recovers most of the lost precision, since resampling the design captures the joint behaviour of the cells. The map from the data to $\bar{\pi}$ involves minima, maxima and a binding constraint that can switch from one arm of (4) to the other, so the functional is directionally but not fully differentiable, and the standard bootstrap is consistent for such a functional only where it is fully differentiable (Fang and Santos, 2019). Its validity must then be assessed by simulation on known truths that preserve the survey design, the cell structure, the belief intervals and the censoring pattern. Appendix B.4 removes that reliance with a Gaussian multiplier form of the numerical delta method of Hong and Li (2018), built on the design's linearization covariance. It states the conditions under which the one-sided bound is asymptotically valid and supplies the lower bounds reported below, with replication as a design-

based cross-check.

5 Data and Belief Measurement

5.1 Data

The application uses the Health and Retirement Study, a biennial panel of Americans over fifty that asks probabilistic expectations questions in a dedicated section of every core interview and follows respondents until death (Health and Retirement Study, 2025; RAND Center for the Study of Aging, 2025). The HRS is sponsored by the National Institute on Aging (grant numbers NIA U01AG009740 and NIA R01AG073289) and is conducted by the University of Michigan. Wave 7, fielded in 2004, is the anchor at which beliefs are elicited, and later waves, the HRS Tracker and the National Death Index supply realizations (Health and Retirement Study, 2026b,a). This analysis uses Early Release data from the 2024 HRS Early Core, which have not been cleaned and may contain errors that will be corrected in the final public release. Of 20,129 respondents interviewed at the anchor wave, 17,983 answer at least one probability question.

The belief is the respondent's own survival probability. Respondents are asked for the percent chance of living to a target age that the instrument adapts to their current age, so the question is individually tailored and not common across respondents. The adapted target is recorded, which makes the realization well defined at the individual level. The event is $Y_i = 1$ if respondent i reaches their own target age.

The realization is constructed from mortality and from evidence that the respondent was alive at a known age. It resolves as one when the respondent died at or after the target age, or was alive at an age at or above it. It resolves as zero when the respondent died before the target age. It is left *unresolved* when the respondent was last known alive below the target. The RAND longitudinal file supplies deaths and interview ages through 2022. Two sources add to it. The HRS Tracker records the last date on which each respondent was known to be alive, from their own interview, a spouse's interview or a field contact, and the 2024 core interview records age in 2024. The National Death Index, linked by the HRS and verified by its staff, records year and quarter of death for respondents the RAND file leaves unresolved, including many who left the panel without an exit interview. Appendix H gives the rules. Together they resolve 1,027 outcomes the RAND file leaves open and never overwrite an outcome it records. Because the targets are set well beyond the respondent's age at interview, the unresolved group remains large, at 30.5 percent of the estimation sample and 39.4 percent of its weight, against 38.1 and 47.4 percent under the RAND file alone. Unresolved cases are never dropped, since dropping them would select on the outcome. They enter as mass allocated in the way most favourable to rationality, following Proposition 5.

Estimation uses the complex survey design. Standard-error strata and half-sample codes give 56 strata with two half-samples each, and person-level analysis weights are used for all population quantities. The weights represent community-dwelling older Americans in 2004, so institutionalized respondents, who carry a separate nursing-home weight, are outside the estimand. Age-ineligible

younger spouses carry zero weight, 817 respondents with a mean age of 47.7 against 65.5 among the weighted, leaving 12,745 of 13,562, or 94.0 percent, with a positive weight. The estimation sample also excludes respondents who report only on the focal grid, meaning that every probability they give is zero, fifty or one hundred, for the reason given below. The estimand is the community-dwelling older population of 2004 *among those who report probabilities on a grid finer than the focal one.*

Table 1: Estimation sample, belief measurement and realization resolution

Panel A. Sample	
Respondents interviewed at the anchor wave	20,129
answering at least one probability question	17,983
in the authoritative estimation sample	13,562
with positive respondent weight	12,745
Households represented	9,939
Share in a household with two or more sample members (%)	53.0
Sampling-error strata	56
Finest-partition cells (atoms)	244
Panel B. Reporting granularity, of 17,983 with a classifiable grid	
Grid 1, some answer not a multiple of five	1,045 (5.8%)
Grid 5, multiples of five	5,318 (29.6%)
Grid 10, multiples of ten	7,600 (42.3%)
Grid 50, focal $\{0, 50, 100\}$ only	4,020 (22.4%)
Mean probability questions used per respondent	6.0
Survival-belief respondents meeting all other criteria	16,780
focal-only, excluded, status not identified	3,218 (19.2%)
Panel C. Survival realization	
Unresolved at last observation (%)	30.5
Weighted rate of reaching the target age, resolved cases	0.4613
Zero-weight respondents	817
mean age	47.7
mean age among positively weighted respondents	65.5

Notes. Health and Retirement Study, Wave 7 (2004), with realizations from Waves 8 through 16 (Health and Retirement Study, 2025; RAND Center for the Study of Aging, 2025). Panel B shares are of the 17,983 respondents with a classifiable reporting grid, which is the finest granularity a respondent displays across all subjective-probability questions answered at the anchor wave. The last two rows of Panel B use as denominator the respondents who report a survival belief and meet every other sample criterion, of whom the focal-only group is then excluded because their point-belief rational-expectations status is not identified. A realization is unresolved when the respondent is alive at last observation and has not yet reached their own target age. Unresolved cases are retained and their mass is allocated as in Proposition 5. Weighted quantities use the person-level respondent weight and describe the population that weight represents, among respondents whose reporting grid is finer than the focal one.

5.2 Reported probabilities

Subjective probabilities in household surveys are reported coarsely, with mass at zero, fifty and one hundred. A reported zero taken literally imposes an absolute restriction, since a rational agent reporting zero places no mass on the event and a single occurrence in such a cell is then unrationalizable.

The rule is fixed in advance and treats a respondent's finest observed granularity as the grid on which they report. Using every subjective-probability question the respondent answered at the anchor wave, on average 6.0 of them, the reporting grid g_i is set to one if any answer is not a multiple of five, to five if any answer is a multiple of five but not of ten, to ten if any answer is a multiple of ten but not focal, and to fifty if every answer lies in $\{0, 50, 100\}$. A report x_i then becomes the interval $[x_i - g_i/2,\ x_i + g_i/2]$ truncated to the unit interval. A zero from a respondent who elsewhere reports thirty-seven and sixty-three stays within half a percentage point of zero, while a zero from a respondent who never leaves the focal grid becomes [0, 0.25]. Only 5.8 percent of respondents report at a resolution finer than multiples of five, and 22.4 percent never leave $\{0, 50, 100\}$.

The rule identifies how coarsely a respondent reports. It does not identify whether a focal-only respondent holds a precise point belief and rounds it, or holds an imprecise belief that no point summarizes. The first is measurement error and is handled by the interval bound of Proposition 4. Under the second the benchmark $p = \mathbb{P}(Y = 1 \mid I)$ is not defined for that respondent, and the point-belief version of rational expectations has no testable content. The authoritative specification therefore excludes focal-only respondents from point-belief inference and reports them separately as a group whose rational-expectations status is *not identified*, instead of counting them as rationalizable by default. Of the 16,780 respondents who report a survival belief and meet every other sample criterion, 3,218, or 19.2 percent, report only on the focal grid and are excluded on this ground. Specifications that retain them with wide intervals, and specifications that take all reports literally, are treated as robustness.

The cell bound of Section 3 is sharp for a cell defined by a single belief interval, but the estimation cells group respondents whose own intervals differ, and a cell is assigned the smallest interval containing its members' intervals. The maximization inside a cell may then select a belief that no member admits. That relaxation can only raise the rationalizable share, so the reported values of $\bar{\pi}$ are upper bounds on the sharp bound computed with respondent-specific intervals. Grouping by the report relies on (1) with $T_i = r_i$.

6 Empirical Results

6.1 Rejecting rational expectations

Across 47 cells and 94 one-sided hypotheses, the exact test rejects the hypothesis that the whole population satisfies the rational-expectations restriction by a margin far beyond any conventional

level (Table 2). Under the maintained measurement and information assumptions this rejects rational expectations for the population as a whole, without identifying any individual whose belief was irrational. The rejection stands when one respondent per household is retained, and under a household-clustered bootstrap that relaxes the within-cell independence the binomial requires, although the bootstrap costs roughly thirty orders of magnitude in the adjusted p-value. It also stands when the same inequalities are evaluated with standard errors from balanced repeated replication, which permits dependence within the sampling-error half-samples. A simulation under full rational expectations that preserves the observed cell sizes and censoring rates rejects at rate 0.000, so the test is conservative in this setting.

Table 2: Exact test of the rational-expectations restriction

Panel A. Inference routes

Route	Cells	Tests	Smallest Holm p	Decision at 1%
Exact binomial	47	94	4.3×10^{-58}	reject
One respondent per household	43	86	8.4×10^{-58}	reject
Household-clustered bootstrap	47	94	2.1×10^{-27}	reject
Half-sample replication	47	94	2.4×10^{-34}	reject

Panel B. Most violating cells

Belief interval	n	Reached / unresolved	Direction of violation
[0.97, 1.00]	203	64 / 68	too few possible survivors
[0.95, 1.00]	487	186 / 167	too few possible survivors
[0.00, 0.03]	322	57 / 79	too many survivors
[0.00, 0.05]	696	122 / 177	too many survivors

Notes. Each cell contributes the two one-sided exact binomial hypotheses of equation (9), and Holm's step-down procedure is applied to the full collection, which controls the family-wise error rate in finite samples under arbitrary dependence across tests. Cells with fewer than twenty-five respondents are not tested, which leaves the global test valid and can only cost power. Exactness of an individual p-value also requires independent observations within a cell, which the second and third rows of Panel A relax by retaining one respondent per household and by resampling households. The fourth row replaces the binomial with a one-sided normal approximation whose standard errors come from balanced repeated replication over the sampling-error strata and half-samples, so dependence is permitted at the level of the sampling-error half-samples. That route is design-aware and not exact, and the exact finite-sample statement belongs to the binomial row alone. A simulation in which the restriction holds for everyone, preserving the observed cell sizes and censoring rates rejects at rate 0.000 at the ten, five and one percent levels. "Too many survivors" means the realized rate exceeds any belief the interval admits. "Too few possible survivors" means that even counting every unresolved case as a survivor the cell cannot reach the lowest admissible belief.

The violations run in both directions and sit where the restrictions bind, at beliefs far from one half. Respondents whose reporting interval reaches no higher than three or five percent are already known to have reached their target age at rates near eighteen percent. Respondents whose interval starts at ninety-five percent or above are known to have reached it at rates between thirty-two and thirty-eight percent, and even counting every unresolved case as a survivor they cannot approach the lowest belief their own interval admits. People nearly certain they will die before the target age outlive it far too often, and people nearly certain they will survive die far too often (Figure 6

in Appendix F shows every cell). These cells establish that the universal hypothesis fails and carry no information about how much of the population is implicated.

6.2 How many could have rational expectations?

The ladder is a nested sequence of asserted information sets $Z_0 \subseteq Z_1 \subseteq \cdots \subseteq Z_4$, and each set is a rung. Z_0 is the reported belief alone, and each later rung adds one of the respondent's own sex, age group, self-rated health and current smoking, in that order. The cells of every rung are unions of a single finest partition of the sample into 244 atoms, where an atom is a group of respondents who share the same belief group, the reported probability rounded to the nearest tenth, and the same values of all four variables.

Conditioning on the reported belief alone, the maximum rationalizable share is $\bar{\pi} = 0.9671$, so the minimum non-rationalizable share is 3.29 percent of the weighted population, with a one-sided 95 percent lower bound of 2.79 percent from the multiplier procedure of Appendix B.4 and 2.85 from balanced repeated replication. Under the weakest assertion about what agents know, the data are consistent with roughly ninety-seven in every hundred people belonging to a component that satisfies the restriction. Under the RAND file alone, which the outcome coding of Section 5 extends, the same share is 2.24 percent with a lower bound of 1.82, so the lower bound with the fuller record exceeds the point estimate the RAND file allows. Section 6.3 traces the difference.

The estimate is computed among respondents whose point-belief status is defined, which excludes those who report only on the focal grid, whose point-belief status is not identified. Extending the statement to everyone eligible uses an additional assumption, that the reporting condition also holds with respect to focal status f, a function of the reporting grid, so that

$$\mathbb{E}_{P^T_{\mathrm{RE}}}[\, Y \mid p, Z, r, f \,] = p.$$

This is stronger than the condition with $T = r$ used for the estimates above and does not follow from it. Under it the two groups occupy separate cells, and by Proposition 3, writing S for the weighted share whose status is identified, $1 - \bar{\pi}_{\mathrm{all}} = S(1 - \bar{\pi}_{\mathrm{identified}}) + (1 - S)(1 - \bar{\pi}_{\mathrm{focal}})$. Within the identified group these cells coincide with the estimation cells, so $\bar{\pi}_{\mathrm{identified}}$ is the estimate above. Giving the focal group the most favourable treatment possible, $\bar{\pi}_{\mathrm{focal}} \leq 1$, yields the conservative bound

$$1 - \bar{\pi}_{\mathrm{all}} \;\geq\; S\left(1 - \bar{\pi}_{\mathrm{identified}}\right). \tag{12}$$

The assumption is sufficient for this decomposition, and the inequality may hold in particular cases without it. Here $S = 0.834$, so the minimum non-rationalizable share for the full eligible population is at least 2.74 percent at the base rung with a one-sided lower bound of 2.32, and 6.91 percent at the finest with a lower bound of 5.38. The inference treats the product directly, and replication gives 2.38 and 6.01 for the same two objects. The first statement concerns the respondents whose reports admit a point belief, and the second concerns everyone eligible.

Requiring beliefs to be efficient with respect to information the respondent unambiguously

possesses moves the minimum non-rationalizable share from 3.29 percent to 8.28 percent, and the lower confidence bound from 2.79 percent to 6.45 percent, which more than doubles (Table 3 and Figure 2). At the point estimate for that rung, no decomposition consistent with the data can assign more than 91.72 percent of the population to a component satisfying the restriction, even with the rest left unrestricted. This does not single out a group of individuals whose forecasts are irrational. The respondent's own sex moves the bound by a tenth of a point, from 0.9671 to 0.9661. Their age group produces the largest single increase, taking the non-rationalizable share from 3.39 to 6.35 percent. Self-rated health raises it to 7.58 percent and current smoking to 8.28 percent.

The bound does not move with the predictive content of a conditioning variable. Adding Z lowers $\bar{\pi}$ only when it uncovers heterogeneity in the compatibility between beliefs and realizations that aggregation over Z had concealed. A variable can predict mortality well and leave the bound untouched, provided beliefs already incorporate it correctly. Age, self-rated health and smoking reveal discrepancies between beliefs and outcomes that coarser cells mask, while sex reveals little at the point where the ladder adds it.

Table 3: Maximum rationalizable share along the information-set ladder

		Weighted				Unweighted
Rung	Information set Z_j	$\bar{\pi}$	$1-\bar{\pi}$	95% LCB	BRR	$\bar{\pi}$
Z_0	belief only	0.9671	0.0329	0.0279	0.0285	0.9503
Z_1	+ own sex	0.9661	0.0339	0.0272	0.0293	0.9464
Z_2	+ own age group	0.9365	0.0635	0.0515	0.0572	0.9153
Z_3	+ own self-rated health	0.9242	0.0758	0.0611	0.0689	0.9009
Z_4	+ own current smoking	0.9172	0.0828	0.0645	0.0716	0.8920

Notes. $\bar{\pi}$ is the maximum share of the population whose beliefs can be rationalized as satisfying rational expectations with respect to the stated information set, and $1-\bar{\pi}$ is the corresponding minimum non-rationalizable share. Each rung conditions on the previous one plus the listed variable, all of which are attributes the respondent unambiguously knows about themselves. The ladder is nested by construction, since a single finest partition of 244 atoms is fixed in advance and every coarser rung is obtained by aggregating those same atoms, so the monotonicity of Theorem 1 holds exactly and not approximately. Weighted columns use the respondent weight and describe the population that weight represents among respondents whose reporting grid is finer than the focal one, which is the conditional estimand of Section 6. Lower confidence bounds are one-sided at 95 percent. The certified LCB column comes from the multiplier numerical delta method of Appendix B.4 with the step rule $\kappa_H = H^{1/4} = 2.74$, and the BRR column is the design-based cross-check from balanced repeated replication over the 56 sampling-error strata and their half-samples, using 64 replicates. The rungs are not causal. They are alternative assertions about what the analyst is willing to assume the agent knows.

Because monotonicity holds in the population, the informative quantity is the size of the decline, and reading it requires a benchmark. Refining cells depresses a plug-in bound at a kink, and the belief intervals turn the kink into the plateau of Proposition 4, which removes most of that bias. A population that is rational by construction, in which each respondent's own reported belief generates their outcome and the censoring pattern is held fixed, returns 0.00 percentage points at the first two rungs, 0.32 at the third, 0.76 at the fourth and 1.34 at the finest. Under this rational-null benchmark, cell refinement generates a shortfall of 1.34 percentage points at the finest

rung, roughly one sixth of the observed 8.28 percent. The comparison indicates the scale of the finite-sample effect and is not a decomposition of the observed estimate into bias and signal. At the base rung the benchmark is 0.00. Under literal beliefs the benchmark is far larger (Appendix F).

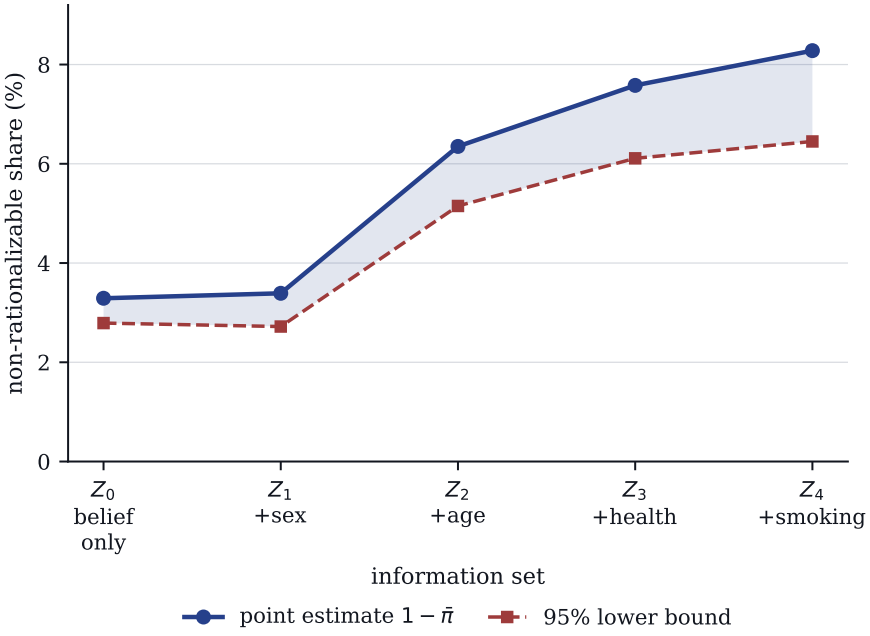


Figure 2: The minimum non-rationalizable share along the information-set ladder

Notes. Weighted estimates of $1-\bar{\pi}$ in percent, with one-sided 95 percent lower confidence bounds from the multiplier numerical delta method of Appendix B.4. The replication bounds, which sit at or slightly above these, are in Table 3. Each rung conditions on the previous one plus the listed attribute. The ladder is nested, so the decline in $\bar{\pi}$ follows from Theorem 1. A population that is rational by construction, on the same cells and with the same censoring, returns 0.00 percentage points at the first two rungs rising to 1.34 at the finest, so part of the movement is a finite-sample consequence of refining cells.

Age matters most whatever the order in which the variables enter. Over all twenty-four orderings it contributes between 2.77 and 3.67 points of the 5.00-point rise, and sex contributes least in every ordering (Appendix F).

Formed inside each replicate, the rise on adding age group is 2.96 points with a one-sided lower bound of 2.42, on adding self-rated health 1.24 with a lower bound of 0.84, and on adding smoking 0.70 with a lower bound of 0.12. Sex adds 0.10, with a lower bound of 0.00. The increments are a different object from the levels, and on the coverage designs of Appendix B only the age increment holds the nominal level, with a minimum coverage of 0.967 against 0.923, 0.913 and 0.743 for sex, health and smoking. Conditioning on age therefore reveals at least 2.42 additional percentage points of non-rationalizability, and the other three increments are reported as replication intervals.

Figure 3 sets each increment against what refinement alone would produce. Permuting the added variable within the cells of the previous rung preserves its number of categories, its marginal distribution and the resulting cell sizes, and destroys only its relation to beliefs and outcomes.

Across 1,000 permutations the median increase is 0.12 points for age, 0.29 for health and 0.31 for smoking, so refinement alone moves the bound. The ninety-ninth percentile of the permutation draws is 0.28, 0.53 and 0.53 points, and no draw attains the observed increment for any of the three variables. Permuting sex leaves the bound almost unchanged, with a ninety-ninth percentile of 0.02 points, and one draw in a thousand reaches the observed 0.10. Sex therefore carries a small conditional discrepancy that refinement alone does not produce, too small to separate from sampling error.

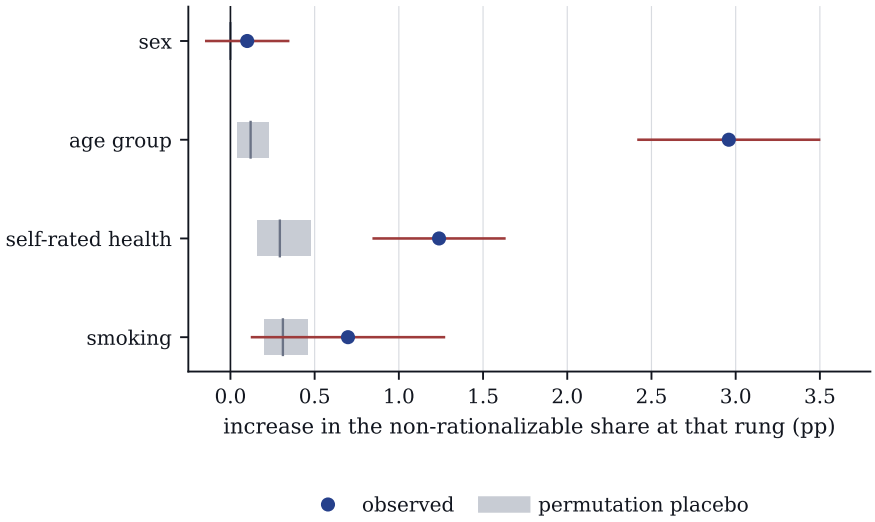


Figure 3: Observed information content against mechanical refinement

Notes. Points are the increase in the weighted minimum non-rationalizable share when the listed variable is added to the previous rung, in percentage points, with whiskers spanning the point estimate plus and minus 1.645 replication standard errors, which are nominal one-sided 95 percent normal-approximation intervals. Coverage diagnostics for the individual increments are reported in the text. Grey bands run from the fifth to the ninety-fifth percentile of 1,000 permutations in which the variable is shuffled among respondents within the cells of the previous rung, which holds the number of categories, the marginal distribution and the cell sizes fixed while removing any relation to beliefs and outcomes. The tick inside each band is the permutation median. Increments are computed from unrounded levels.

The shortfall at the base rung is the work of a handful of cells at the tails of the belief distribution, and removing the five largest contributors eliminates it. At the finest rung 173 of 244 cells contribute and removing the ten largest still leaves 5.95 of the 8.28 percentage points, so what the ladder adds is dispersed across many cells.

Keying cells on each respondent's own belief interval, which is closer to Proposition 4, raises the estimate to 4.45 percent at the base rung and 11.51 at the finest. Those cells have a median of 4 respondents, and on cells that small the plug-in bound does not support reliable inference, as Appendix F shows.

The reporting assumption (1) also moves the level. Requiring it for the report alone, with intervals that differ within a report, the sharp minimum non-rationalizable share is 4.35 percent at the base rung and 10.16 percent at the finest, so the reported 3.29 and 8.28 percent are conservative

under that assumption. Dropping (1), respondents from different report groups who share a latent belief may pool, and the sharp share solves a linear program over where each latent belief lies within its own interval. It is then at most 2.68 percent at the base rung and 7.50 percent at the finest. These are population bounds evaluated on the estimation sample, without inference. How much of the shortfall the data establish depends on whether report groups may offset one another.

The belief-measurement model moves the level of the bound and leaves the pattern intact. Taking reports literally gives the tightest bound, since a reported zero or one hundred then imposes an absolute restriction. Restricting to respondents who report on a fine grid with no interval widening gives nearly the same answer as giving every respondent their own rounding interval, which points to a real difference between precise and coarse reporters (Appendix F).

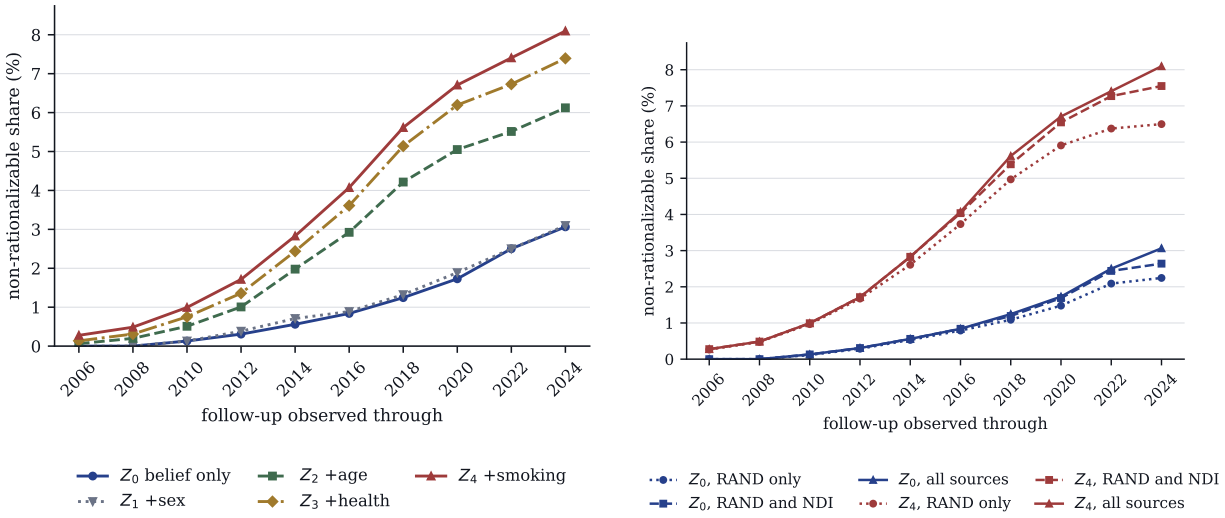
Weighting also moves the level and leaves the pattern unchanged. The unweighted ladder in the final column of Table 3 runs from 0.9503 to 0.8920, against 0.9671 to 0.9172 weighted, so weighting moves the bound toward less violation at every rung. The age-ineligible younger spouses who carry zero weight contribute to that difference, and the ordering across rungs is identical in the two columns.

The confidence bounds of Table 3 come from the multiplier numerical delta method of Appendix B.4, which carries the asymptotic justification of Proposition 8. In simulations that hold the design, weights and censoring fixed and place the true shortfall at the boundary, in the interior, in a single binding cell and at the non-differentiable point, its minimum coverage is 0.990 against a nominal 0.95. Balanced repeated replication gives slightly higher bounds at every rung and covers at 0.982 at the empirical value of the finest rung. These simulations assess coverage on the designs they contain, near the configurations the data come close to, and Appendix B reports them.

6.3 Identification through follow-up

By Proposition 6, every share above is a lower bound on what longer follow-up of the same people would show, and the data allow that second dimension to be measured. Holding the 2004 cohort, its beliefs, cells and weights fixed, the outcome can be recomputed using only the evidence dated by a given year. The series is reconstructed from files released since, so each point describes the evidence about events up to that year and not the files an analyst could have held at the time, since death records and tracking contacts reach the public data with a lag. Figure 4 reports the result. In 2010, with 88.5 percent of the weight still unresolved, the minimum non-rationalizable share is 0.13 percent at the base rung and 0.99 at the finest. By 2016 it is 0.84 and 4.07, by 2022 2.50 and 7.41, and with evidence through 2024 3.06 and 8.10. The headline values add the 2024 interviews that were conducted in 2025, 204 of which confirm that a respondent reached the target. These 204 people carry 1.93 percent of the weight and account for the whole distance between the last point of the figure and Table 3. Neither curve has flattened, and the distance between the base rung and the finest widens as outcomes resolve.

Rejecting rational expectations needs little follow-up. Applied to the outcome coded with the evidence dated by 2008, when 91.4 percent of the sample is still unresolved and the finest-rung



(a) All sources, each information set

(b) Sources of follow-up, Z_0 and Z_4

Figure 4: The minimum non-rationalizable share as follow-up of the 2004 cohort lengthens

Notes. Weighted point values of $1-\bar{\pi}$ in percent when only the evidence dated by the year on the horizontal axis is used, with the 2004 beliefs, belief intervals, cells and weights held fixed. Panel (a) uses every source. Panel (b) uses the RAND file alone, adds National Death Index deaths, and adds alive evidence from the HRS Tracker and the 2024 core interview. Deaths that conflict with another source never resolve an outcome at any date. Each point is reconstructed from files released later. The last point uses evidence through 2024. The headline values of Table 3 also use the 2024 core interviews conducted in 2025 and the Tracker contacts recorded in 2025.

share is 0.49 percent, the exact test already rejects, with a smallest Holm-adjusted p-value of 5.7×10^{-4}. By 2010 that p-value is 7.5×10^{-9}. A few cells at the tails of the belief distribution fail as soon as their outcomes begin to resolve, while the share of people who cannot be reconciled with the restriction accumulates cell by cell as the forecast future arrives, to 8.28 percent at the finest rung with all current sources. How much of a population is compatible with rational expectations depends on what the analyst has observed as well as on what agents are assumed to know, and both move the answer in the same direction. A study that stops early, or a panel that loses track of its respondents, understates the incompatible share.

Part of the rise comes from information about the same years that the RAND file does not carry. The RAND file alone gives 2.24 percent at the base rung and 6.50 at the finest. Deaths recorded by the National Death Index, many of them among respondents who left the panel without an exit interview, and survivals confirmed by the 2024 interview each raise the share on their own, and together they give the headline 3.29 and 8.28. In the retrospective series the codings coincide through 2016 and separate afterwards (Figure 4, panel b). Appendix H separates the sources and reports the rules, the conflicts between sources and the sensitivity of the headline to them.

6.4 Rationalizability across forecast domains

The same instrument asks respondents who have not yet reached sixty-two for the percent chance that they will be working full time after that age, and the measurement architecture applies to it without modification. The outcome is full-time employment at the first interview at or after the target age, with death before it counted as not working, which leaves only 13.7 percent unresolved against 30.5 percent for survival. 4,078 respondents qualify and 3,476 carry a positive weight.

The rational-expectations restriction is rejected for the whole population in this domain too, over 28 cells and 56 one-sided hypotheses. The identified magnitude is far larger than for survival. Conditioning on the reported belief alone, the minimum non-rationalizable share is 11.54 percent with a one-sided 95 percent lower bound of 10.00 percent, against 3.29 percent for survival at the same rung. A rational population returns 0.01 percentage points on the same cells, so the estimate is not a small-sample artifact. Beliefs about working late in life are harder to reconcile with what happens than beliefs about living.

Figure 5 applies the base rung to every personal probability in the module whose realization the panel resolves cleanly, five domains in all. The two survival questions give the smallest shortfalls, 3.29 percent for the adapted target age and 1.78 for reaching seventy-five. Nursing-home entry within five years gives 4.17. The two employment questions give the largest, 8.32 percent for working after sixty-five and 11.54 for working after sixty-two. The rational-null benchmark does not exceed 0.01 percentage points in any of them, so the spread is not a cell-size effect.

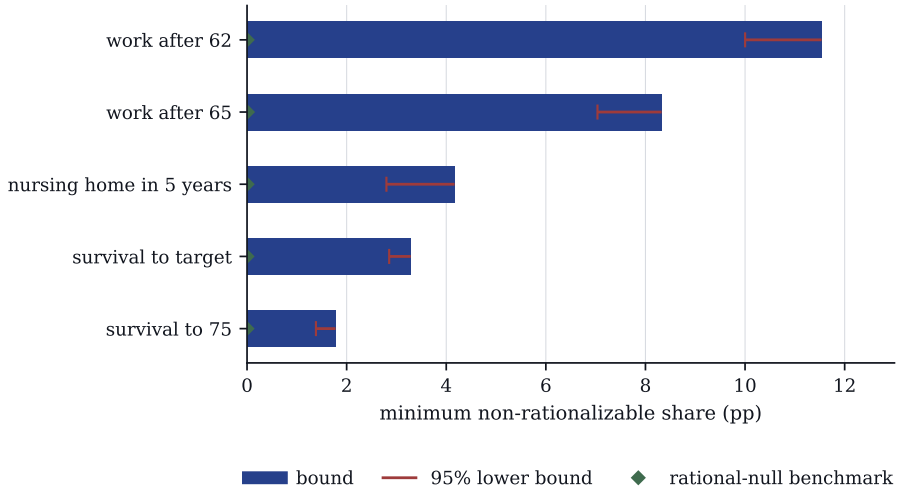


Figure 5: The minimum non-rationalizable share across expectation domains

Notes. Weighted estimates of $1 - \bar{\pi}$ conditioning on the reported belief alone, in percentage points, with the whisker running to the one-sided 95 percent lower bound from balanced repeated replication. Diamonds mark the shortfall a population that is rational by construction produces on the same cells. Each domain uses the measurement architecture of the survival application without modification and its own eligible sample. Survival to the target age uses the outcome coding of Section 5, and the other domains use the RAND file. Bequest and inheritance items are excluded because the panel does not resolve them for surviving respondents.

Coverage of the lower bounds in Figure 5 at the boundary, where the truth is zero, is 1.000 in every domain. At an interior design calibrated to a shortfall of five percent, coverage is 0.975 and 0.980 for the two survival questions, 0.950 for nursing-home entry, and 0.900 and 0.920 for the two employment questions. The two employment bounds fall short of the nominal 0.95 at that design and are reported as replication intervals.

Each domain is asked of a different set of respondents over a different horizon, so the atlas compares samples as well as domains. Estimating each domain and survival on exactly the same respondents separates the two (Table 4). Among the 3,397 respondents eligible for the work-after-sixty-two question, that domain gives 11.38 percent while survival gives 0.45 on the same people, so the gap is wider on a common sample than in the atlas. Among the 6,243 respondents eligible for the nursing-home question the ordering reverses, with nursing home at 4.56 percent and survival at 13.82.

The survival bound itself runs from 0.45 to 13.82 percent across these subsamples, against 3.29 in the full one. The maximum rationalizable share depends on the domain and on the population, and the work-versus-survival contrast is partly a horizon effect, since survival outcomes are mostly unresolved among respondents under sixty-two.

Table 4: Each domain and survival, estimated on the same respondents

Domain and its eligible sample	n	Domain	Survival
Survival to 75	6,781	1.80	1.16
Nursing-home entry in five years	6,243	4.56	13.82
Work full time after 65	4,077	8.33	0.65
Work full time after 62	3,397	11.38	0.45

Notes. Weighted minimum non-rationalizable share $1 - \bar{\pi}$ in percent, conditioning on the reported belief alone. Each row restricts to the respondents eligible for that domain who also report a survival belief on a non-focal grid, and estimates both bounds on exactly those people with the same architecture and the same weights. The survival column therefore varies across rows because the sample does. The rational-null benchmark is at or below 0.01 percentage points in every cell of the table.

The ladder is shorter in the work domain. That sample is under a third the size of the survival sample, the finest partition holds 232 atoms with a median of six respondents, and the rational-null benchmark rises from 0.01 percentage points at the base rung to 6.33 at the finest, against an estimate of 16.45. Only the base rungs are reported.

Two beliefs elicited from the same respondents also support a joint question, whether the same people can be rationalized in both domains at once. Appendix G gives the joint program a closed form for two domains and shows that the joint restriction can cut a bound far below what the two marginal bounds imply. Applied to survival and work after sixty-two it adds 0.34 percentage points beyond the marginal envelope, so in these data the marginal bounds carry nearly all of the joint content.

7 Conclusion

Rejecting rational expectations and determining how much of a population is incompatible with them are different questions with different answers. The first reduces to a finite collection of one-sided binomial inequalities, and in the Health and Retirement Study a few cells at the tails of the belief distribution settle it by a wide margin. They settle it with evidence dated by 2008, when more than nine outcomes in ten were still unresolved. The second requires aggregating partial information across every cell and every cell weight, and its answer is much smaller. Among older Americans in 2004 who report probabilities on a grid finer than the focal one, at least 2.79 percent of the population mass is incompatible with the calibration restriction on survival given the reported belief, at ninety-five percent confidence. Once beliefs must also be efficient with respect to own sex, age, self-rated health and smoking, that floor is 6.45 percent, more than six people in a hundred.

That share depends on what the analyst assumes agents know and on how much of the forecast future has been observed, and both move it in the same direction. A researcher willing to assert more, that agents know their own diagnosed conditions or their parents' longevity, would obtain a share at least as large, and longer follow-up of the same cohort cannot lower the share reported here. The share also differs across forecast domains and across the populations that different questions define, so there is no single rationalizable fraction to report.

What the bound leaves open matters beyond this application. Rationalizability says which models the observed distribution can be reconciled with, and not how many agents are of a given type, so the identified set for that share runs from zero to the bound however large the sample grows. Any question that depends on how people would revise beliefs in an environment the data do not contain therefore needs more than compatibility with rational expectations in the environment they do. The framework makes the size of that gap measurable, which one needs before assuming it away.

References

Andrews, I., Roth, J., and Pakes, A. (2023). Inference for linear conditional moment inequalities. *Review of Economic Studies*, 90(6):2763–2791.

Bai, Y., Ponomarev, K., Santos, A., Shaikh, A. M., Tabord-Meehan, M., and Torgovitsky, A. (2026). Inference for linear systems with unknown coefficients. Working paper, arXiv:2604.24904.

Bissonnette, L. and de Bresser, J. (2018). Eliciting subjective survival curves: Lessons from partial identification. *Journal of Business and Economic Statistics*, 36(3):505–515.

Bonhomme, S., Jochmans, K., and Robin, J.-M. (2016). Non-parametric estimation of finite mixtures from repeated measurements. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 78(1):211–229.

Bonhomme, S., Lamadon, T., and Manresa, E. (2022). Discretizing unobserved heterogeneity. *Econometrica*, 90(2):625–643.

Bonhomme, S. and Manresa, E. (2015). Grouped patterns of heterogeneity in panel data. *Econometrica*, 83(3):1147–1184.

Bordalo, P., Gennaioli, N., Ma, Y., and Shleifer, A. (2020). Overreaction in macroeconomic expectations. *American Economic Review*, 110(9):2748–2782.

Branch, W. A. (2004). The theory of rationally heterogeneous expectations: Evidence from survey data on inflation expectations. *Economic Journal*, 114(497):592–621.

Caliari, D. and Petri, H. (2024). Irrational random utility models. Working paper, arXiv:2403.10208.

Choi, S., Kariv, S., Müller, W., and Silverman, D. (2014). Who is (more) rational? *American Economic Review*, 104(6):1518–1550.

Clopper, C. J. and Pearson, E. S. (1934). The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413.

Coibion, O. and Gorodnichenko, Y. (2015). Information rigidity and the expectations formation process: A simple framework and new facts. *American Economic Review*, 105(8):2644–2678.

Cornea-Madeira, A., Hommes, C., and Massaro, D. (2019). Behavioral heterogeneity in U.S. inflation dynamics. *Journal of Business and Economic Statistics*, 37(2):288–300.

Cunha, F., Heckman, J. J., and Navarro, S. (2005). Separating uncertainty from heterogeneity in life cycle earnings. *Oxford Economic Papers*, 57(2):191–261.

D'Acunto, F., Hoang, D., Paloviita, M., and Weber, M. (2023). IQ, expectations, and choice. *Review of Economic Studies*, 90(5):2292–2325.

Das, M., Dominitz, J., and van Soest, A. (1999). Comparing predictions and outcomes: Theory and application to income changes. *Journal of the American Statistical Association*, 94(445):75–85.

D'Haultfœuille, X., Gaillac, C., and Maurel, A. (2021). Rationalizing rational expectations: Characterizations and tests. *Quantitative Economics*, 12(3):817–842.

Elliott, G., Komunjer, I., and Timmermann, A. (2005). Estimation and testing of forecast rationality under flexible loss. *Review of Economic Studies*, 72(4):1107–1125.

Fang, Z. and Santos, A. (2019). Inference on directionally differentiable functions. *Review of Economic Studies*, 86(1):377–412.

Fang, Z., Santos, A., Shaikh, A. M., and Torgovitsky, A. (2023). Inference for large-scale linear systems with known coefficients. *Econometrica*, 91(1):299–327.

Giustinelli, P., Manski, C. F., and Molinari, F. (2022). Precise or imprecise probabilities? evidence from survey response related to late-onset dementia. *Journal of the European Economic Association*, 20(1):187–221.

Goff, L. and Mbakop, E. (2025). Testing the solvability of systems of linear inequalities. Working paper, arXiv:2506.06776.

Health and Retirement Study (2025). Health and retirement study, (RAND HRS longitudinal file 2022 (v1)) public use dataset. Produced and distributed by the University of Michigan with funding from the National Institute on Aging (grant numbers NIA U01AG009740 and NIA R01AG073289), Ann Arbor, MI. May 2025.

Health and Retirement Study (2026a). 2024 HRS early core public use dataset, version 1.0. Produced and distributed by the University of Michigan with funding from the National Institute on Aging (grant number NIA U01AG009740), Ann Arbor, MI. September 2026.

Health and Retirement Study (2026b). Health and retirement study 2022 tracker, version 2.0. Produced and distributed by the University of Michigan with funding from the National Institute on Aging (grant number NIA U01AG009740), Ann Arbor, MI. October 2026.

Heeringa, S. G. and Connor, J. H. (1995). Technical description of the Health and Retirement Survey sample design. HRS Documentation Report DR-002, Institute for Social Research, University of Michigan, Ann Arbor, MI.

Henry, M., Kitamura, Y., and Salanié, B. (2014). Partial identification of finite mixtures in econometric models. *Quantitative Economics*, 5(1):123–144.

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.

Hong, H. and Li, J. (2018). The numerical delta method. *Journal of Econometrics*, 206(2):379–394.

Horowitz, J. L. and Manski, C. F. (1995). Identification and robustness with contaminated and corrupted data. *Econometrica*, 63(2):281–302.

Hurd, M. D. (2009). Subjective probabilities in household surveys. *Annual Review of Economics*, 1:543–562.

Hurd, M. D. and McGarry, K. (2002). The predictive validity of subjective probabilities of survival. *Economic Journal*, 112(482):966–985.

Keane, M. P. and Runkle, D. E. (1990). Testing the rationality of price forecasts: New evidence from panel data. *American Economic Review*, 80(4):714–735.

Kitamura, Y. and Stoye, J. (2018). Nonparametric analysis of random utility models. *Econometrica*, 86(6):1883–1909.

Krewski, D. and Rao, J. N. K. (1981). Inference from stratified samples: Properties of the linearization, jackknife and balanced repeated replication methods. *Annals of Statistics*, 9(5):1010–1019.

Manski, C. F. (2004). Measuring expectations. *Econometrica*, 72(5):1329–1376.

Manski, C. F. and Molinari, F. (2010). Rounding probabilistic expectations in surveys. *Journal of Business and Economic Statistics*, 28(2):219–231.

Mellers, B., Stone, E., Murray, T., Minster, A., Rohrbaugh, N., Bishop, M., Chen, E., Baker, J., Hou, Y., Horowitz, M., Ungar, L., and Tetlock, P. E. (2015). Identifying and cultivating superforecasters as a method of improving probabilistic predictions. *Perspectives on Psychological Science*, 10(3):267–281.

Mogstad, M., Santos, A., and Torgovitsky, A. (2018). Using instrumental variables for inference about policy relevant treatment parameters. *Econometrica*, 86(5):1589–1619.

Muth, J. F. (1961). Rational expectations and the theory of price movements. *Econometrica*, 29(3):315–335.

RAND Center for the Study of Aging (2025). RAND HRS longitudinal file 2022 (v1). Produced by the RAND Center for the Study of Aging, with funding from the National Institute on Aging and the Social Security Administration, Santa Monica, CA. May 2025.

Rudas, T., Clogg, C. C., and Lindsay, B. G. (1994). A new index of fit based on mixture methods for the analysis of contingency tables. *Journal of the Royal Statistical Society, Series B (Methodological)*, 56(4):623–639.

Souleles, N. S. (2004). Expectations, heterogeneous forecast errors, and consumption: Micro evidence from the michigan consumer sentiment surveys. *Journal of Money, Credit and Banking*, 36(1):39–72.

Stango, V. and Zinman, J. (2020). Behavioral biases are temporally stable. Working Paper 27860, National Bureau of Economic Research.

Stango, V. and Zinman, J. (2023). We are all behavioural, more, or less: A taxonomy of consumer decision-making. *Review of Economic Studies*, 90(3):1470–1498.

Torgovitsky, A. (2019a). Nonparametric inference on state dependence in unemployment. *Econometrica*, 87(5):1475–1505.

Torgovitsky, A. (2019b). Partial identification by extending subdistributions. *Quantitative Economics*, 10(1):105–144.

Wolter, K. M. (2007). *Introduction to Variance Estimation*. Springer, 2nd edition.

A Proofs

Throughout, $x = (p, z)$ indexes a cell, F_X is the distribution of cells, and $m(x) = P_{\text{obs}}(Y = 1 \mid X = x)$.

Proof of Proposition 1

Take $\pi = 0$ and $P_A = P_{\text{obs}}$. Condition (3) holds trivially and P_A is unrestricted, so 0 belongs to the identified set for every P_{obs}. The set of feasible π is an interval. If π' is feasible with decomposition (P_{RE}, P_A) and $\pi < \pi'$, then π is feasible with the same P_{RE} and with alternative component $\big[(\pi' - \pi)P_{\text{RE}} + (1 - \pi')P_A\big]/(1 - \pi)$, which is a probability measure because $\pi < \pi' \leq 1$. The identified set is therefore $[0, \bar{\pi}]$ with $\bar{\pi}$ the supremum of feasible values. □

Proof of Proposition 2

Within cell x, let λ denote the mass share of the rational component and let $a \in [0, 1]$ denote the mean of Y under the alternative component. The rational component must satisfy $\mathbb{E}[Y \mid x] = p$ by Definition 1, so

$$m \;=\; \lambda p + (1 - \lambda)a.$$

Solving for a gives $a = (m - \lambda p)/(1 - \lambda)$ when $\lambda < 1$. The constraint $a \geq 0$ is equivalent to $\lambda p \leq m$, and $a \leq 1$ is equivalent to $\lambda(1 - p) \leq 1 - m$. Hence $\lambda \leq m/p$ and $\lambda \leq (1 - m)/(1 - p)$, and with $\lambda \leq 1$ this yields $\lambda \leq \bar{\lambda}(p, m)$ as in (4).

Attainment is by construction. Set $\lambda = \bar{\lambda}(p, m)$. If $m \geq p$ take $a = 1$, which satisfies the identity because $\bar{\lambda} = (1 - m)/(1 - p)$ in that case. If $m < p$ take $a = 0$, which satisfies it because $\bar{\lambda} = m/p$. If $m = p$ take $\lambda = 1$. The rational component's conditional law given the cell is any distribution on $\{0, 1\}$ with mean p, which exists. □

Proof of Proposition 3

The restriction (2) is a conditional-mean restriction given X, so it binds cell by cell and imposes no cross-cell constraint. Let $\lambda(x)$ be the rational mass share in cell x. Feasibility of $(\lambda(x))_x$ is equivalent to $\lambda(x) \leq \bar{\lambda}(p(x), m(x))$ for F_X-almost every x, by Proposition 2. The total rational share is $\int \lambda(x)\, dF_X(x)$, which is maximized pointwise, giving (5).

For sharpness, set $\lambda^*(x) = \bar{\lambda}(p(x), m(x))$ and $\bar{\pi} = \int \lambda^* dF_X$. Define P_{RE} by giving cell x the mass $\lambda^*(x)\, dF_X(x)/\bar{\pi}$ and the conditional law on Y constructed in the proof of Proposition 2. Define P_A as the residual, which is a non-negative measure of total mass $1 - \bar{\pi}$ by construction and hence a probability measure after normalization. Then (3) holds with $\pi = \bar{\pi}$ and $P_{\text{RE}} \in \mathcal{R}(Z)$. □

Proof of Proposition 4

With the belief known only to lie in $[a, b]$, the cell bound is $\max_{p \in [a,b]} \bar{\lambda}(p, m)$. As a function of p the map m/p is decreasing and $(1 - m)/(1 - p)$ is increasing, so their minimum is single-peaked with

maximum at $p = m$, where it equals one. If $m \in [a, b]$ the unconstrained maximizer is admissible and the value is one. If $m < a$ the objective is decreasing on $[a, b]$, so the maximum is at $p = a$, where $m/a < 1 \leq (1 - m)/(1 - a)$ and the value is m/a. If $m > b$ the objective is increasing on $[a, b]$, so the maximum is at $p = b$, where $(1 - m)/(1 - b) < 1$ and the value is $(1 - m)/(1 - b)$.

Letting members of the cell hold different beliefs in $[a, b]$ does not raise the bound. If the rational component places mass S_p at belief p, with total S, calibration within each belief gives it success mass $\sum_p S_p\, p$, which lies between aS and bS. Success mass cannot exceed m and failure mass cannot exceed $1 - m$, so $S \leq m/a$ and $S \leq (1 - m)/(1 - b)$, which are the bounds above. □

Proof of Proposition 5

Let m_1 be the known success mass, $m_0 = 1 - m_1 - u$ the known failure mass and u the unresolved mass, which may be split as $u = u_0 + u_1$ with $u_j \geq 0$. Given a belief p, a rational mass S requires success mass Sp drawn from $m_1 + u_1$ and failure mass $S(1 - p)$ drawn from $m_0 + u_0$, so

$$Sp \leq m_1 + u_1, \qquad S(1 - p) \leq m_0 + u_0, \qquad u_0 + u_1 = u.$$

Adding the two inequalities gives $S \leq m_1 + m_0 + u = 1$. If the two can be made to hold with equality simultaneously then $S = 1$, which requires $u_1 = p - m_1 \in [0, u]$, that is $m_1 \leq p \leq m_1 + u$. If $p > m_1 + u$ the binding constraint is the first with $u_1 = u$, giving $S \leq (m_1 + u)/p$. If $p < m_1$ the binding constraint is the second with $u_0 = u$, giving $S \leq (m_0 + u)/(1 - p) = (1 - m_1)/(1 - p)$. Maximizing over $p \in [a, b]$ as in the proof of Proposition 4 yields (7). □

Proof of Theorem 1

Fix p and let $Z \subseteq Z'$. The map $m \mapsto \bar{\lambda}(p, m)$ is the minimum of two affine functions and of the constant one, hence concave on $[0, 1]$. By the tower property $\mathbb{E}[\, m(p, Z') \mid p, Z\,] = m(p, Z)$, so Jensen's inequality for concave functions gives

$$\mathbb{E}\big[\bar{\lambda}\big(p, m(p, Z')\big) \,\big|\, p, Z\big] \;\leq\; \bar{\lambda}\big(p, \mathbb{E}[m(p, Z') \mid p, Z]\big) \;=\; \bar{\lambda}\big(p, m(p, Z)\big).$$

Taking expectations over (p, Z) and applying Proposition 3 at each information set gives $\bar{\pi}(Z') \leq \bar{\pi}(Z)$.

With interval beliefs and censoring, a cell carries an interval $[a, b]$ and outcome shares (m_1, u), and by Proposition 5 its bound is

$$\bar{\lambda}(a, b, m_1, u) = \min\left\{1,\ \frac{m_1 + u}{a},\ \frac{1 - m_1}{1 - b}\right\},$$

with a ratio whose denominator is zero read as $+\infty$. This expression coincides with (7) in each of its three cases. Two properties of this function deliver the result. For fixed $[a, b]$ it is the minimum of a constant and two affine functions of (m_1, u), hence jointly concave in (m_1, u). And it is non-

increasing in a and non-decreasing in b, so shrinking the interval can only lower it. When a cell c is refined into cells c_k with population shares w_k, the outcome shares average back to the parent, $\sum_k w_k\,(m_1(c_k), u(c_k)) = (m_1(c), u(c))$ with $\sum_k w_k = 1$. In the application each cell is assigned the smallest interval containing the intervals of its members, so $[a_{c_k}, b_{c_k}] \subseteq [a_c, b_c]$ for every k. Hence

$$\sum_k w_k\,\bar{\lambda}\big(a_{c_k}, b_{c_k}, m_1(c_k), u(c_k)\big) \;\le\; \sum_k w_k\,\bar{\lambda}\big(a_c, b_c, m_1(c_k), u(c_k)\big) \;\le\; \bar{\lambda}\big(a_c, b_c, m_1(c), u(c)\big),$$

where the first inequality uses the monotonicity in the interval and the second uses concavity and Jensen's inequality. Summing over parent cells gives $\bar{\pi}(Z') \le \bar{\pi}(Z)$. □

Proof of Proposition 7

By Proposition 3, $\bar{\pi} = 1$ if and only if $\bar{\lambda} = 1$ for F_X-almost every cell. By Proposition 5, $\bar{\lambda} = 1$ in a cell if and only if $[a_c, b_c] \cap [m_1(c), m_1(c) + u(c)] \neq \emptyset$, which holds if and only if $m_1(c) \le b_c$ and $m_1(c) + u(c) \ge a_c$. The second is equivalent to $\mathbb{P}(R_0 = 1 \mid c) = 1 - m_1(c) - u(c) \le 1 - a_c$. □

Proofs for the joint-domain program are in Appendix G.

B Inference

B.1 The exact test

By Proposition 7, $\bar{\pi} = 1$ is equivalent to the conjunction over cells of $m_1(c) \le b_c$ and $m_1(c) + u(c) \ge a_c$. Conditional on the cell count n_c, the number of resolved successes k_{1c} is binomial under the null boundary $m_1 = b_c$, and the number of cases compatible with success, $k_{1c} + k_{uc}$, is binomial under the null boundary a_c. The exact one-sided p-values are those in (10). Every tested cell contributes both hypotheses, so the family has exactly $2C$ members. When $b_c = 1$ the upper inequality is vacuous and when $a_c = 0$ the lower one is, and in those cases the component p-value is set to one and the hypothesis stays in the family. Retaining it leaves the family size fixed and can only make the step-down more conservative. Cells with fewer than twenty-five respondents are excluded from the family, the same threshold used for Figure 6. Testing a subset of the cells preserves validity of the global test and reduces power only.

The global null is the intersection of the $2C$ component nulls. Holm's step-down procedure applied to the ordered p-values controls the family-wise error rate at the nominal level in finite samples under arbitrary dependence across the tests (Holm, 1979). Rejection of any component rejects the global null.

Exactness of a component p-value requires independent Bernoulli observations within the cell, which household surveys violate. Appendix F reports the test restricted to one respondent per household, selected by a deterministic identifier rule that is blind to the outcome, and under a bootstrap that resamples households. A design-aware route in Table 2 evaluates the same inequalities with balanced-repeated-replication standard errors over the strata and half-samples, and it is a normal approximation. A simulation in which the restriction holds for everyone that preserves

the observed cell sizes and censoring rates, with the belief placed at the low endpoint, the midpoint and the high endpoint of each interval, rejects at rate 0.000 at the ten, five and one percent levels, so the procedure is conservative in this configuration.

B.2 Inference on the magnitude

The magnitude satisfies (11) with equality when the cells partition the sample and the $\bar{\lambda}_c$ are sharp, so a lower confidence bound requires simultaneous control over every cell proportion and every cell weight.

The exact route places a Clopper–Pearson interval on each cell proportion and each cell weight at level α/K with K the number of intervals (Clopper and Pearson, 1934), and evaluates the bound at the least favourable point of the resulting box. Validity is immediate, since with probability at least $1-\alpha$ the truth lies in the box and on that event the true magnitude is at least the minimized value. It is severely conservative, because bounding each term at its own worst case ignores that the cells cannot be simultaneously extreme, and it is exact under independent sampling for the unweighted empirical distribution, so its target is the unweighted sample. The multiplier procedure of Appendix B.4 is the route for the weighted target and supplies the lower bounds reported in the main text, with balanced repeated replication as the design-based cross-check.

Balanced repeated replication applied to the aggregate is sharper. The map from cell masses to $\bar{\pi}$ involves minima and a constraint that can switch arms, so the functional is directionally but not fully differentiable and standard resampling need not be consistent for it (Fang and Santos, 2019). Agreement between replication and linearization on smooth statistics, reported in Table 5, is necessary, and coverage is assessed directly by simulation.

B.3 Balanced repeated replication

The HRS supplies standard-error strata and, within each, two half-samples designed for replication or for an approximate two-per-stratum linearization. With $H = 56$ strata, a Sylvester Hadamard matrix of order $R = 64$ assigns, in replicate r and stratum h, one half-sample the weight $2w_i$ and the other zero. The variance estimator is

$$\widehat{\mathrm{Var}}(\hat{\theta}) = \frac{1}{R}\sum_{r=1}^{R}\left(\hat{\theta}_r - \hat{\theta}\right)^2, \tag{13}$$

and the one-sided lower bound is $\hat{\theta} - z_{1-\alpha}\,\widehat{\mathrm{se}}$ (Wolter, 2007).

For a stratified design with two units per stratum the linearized variance collapses to $\sum_h (u_{h1} - u_{h2})^2$, where u_{hi} is the unit total of the linearized variable. The replication and linearized estimators agree to three decimal places on all four smooth statistics in Panel A of Table 5.

B.4 A multiplier numerical delta method

Replication is validated only by simulation. The companion procedure here has an asymptotic justification, and the simulations then check its finite-sample performance. It is a multiplier form of the numerical delta method of Hong and Li (2018).

Fix the finest partition and let θ collect, for every atom, the weighted mass, resolved-success mass and unresolved mass, each expressed as a share of the total analysis weight, so that θ is a vector of order one whose sampling error vanishes at the rate of the design. The belief intervals are fixed by the partition, so the magnitude at each rung is a known map $\psi_j(\theta) = 1 - \bar{\pi}(Z_j)$. The map is invariant to a common rescaling of θ, so the implementation may equivalently form the unnormalized weighted totals and their contrasts, which changes nothing to first order.

Lemma 1 (Directional differentiability)**.** *On the set where every cell total is positive, each ψ_j is Lipschitz continuous and Hadamard directionally differentiable.*

Proof. The censored cell bound of Proposition 5 can be written $\bar{\lambda} = \min\{1,\ (m_1 + u)/a,\ (1 - m_1)/(1 - b)\}$, where an arm whose denominator is zero, $a = 0$ for the second or $b = 1$ for the third, is never binding in (7) and is simply omitted from the minimum, so each cell's bound is the minimum of the remaining finitely many terms, all with strictly positive denominators fixed by the partition. The cell shares (m_1, u) are continuously differentiable functions of the cell totals where the totals are positive, and the cell totals are linear in θ. A pointwise minimum of finitely many continuously differentiable functions is locally Lipschitz and Hadamard directionally differentiable, both properties are preserved by composition with continuously differentiable maps and by the chain rule for Hadamard directional derivatives (Fang and Santos, 2019), and ψ_j is a weighted average of such cell bounds whose weights are smooth in θ on the same set. □

Assumption 1 (Design asymptotics)**.** The atom partition is fixed, so the dimension of θ does not grow along the asymptotic sequence, and every cell total entering ψ_j is bounded away from zero. There is a rate $r_n \to \infty$ with $r_n(\hat{\theta} - \theta) \to_d N(0, \Sigma)$, and the linearization estimator $\widehat{\Sigma} = \sum_h d_h d_h'$, with d_h the half-sample contrast of the totals in stratum h, satisfies $r_n^2\,\widehat{\Sigma} \to_p \Sigma$.

Assumption 1 is the standard first-order theory for stratified multistage designs with two half-samples per stratum, in which the number of strata grows (Krewski and Rao, 1981; Wolter, 2007). It concerns only the linear statistics $\hat{\theta}$, where nothing non-smooth has yet happened.

The procedure draws Gaussian multipliers instead of treating the finite set of replicates as a bootstrap distribution. With $g_{s1}, \ldots, g_{sH}$ independent standard normals,

$$Z_s^* \;=\; \sum_h g_{sh}\, d_h, \qquad D_s \;=\; \frac{\psi\big(\hat{\theta} + \kappa_n Z_s^*\big) - \psi(\hat{\theta})}{\kappa_n}, \tag{14}$$

so that conditional on the data $Z_s^* \sim N(0, \widehat{\Sigma})$, and the one-sided bound is

$$\mathrm{LCB} \;=\; \psi(\hat{\theta}) \;-\; q_{1-\alpha}(D), \tag{15}$$

where $q_{1-\alpha}(D)$ is the empirical $(1-\alpha)$ quantile of the draws D_s themselves, not of $r_n D_s$. The rate never needs to be computed, because the unscaled draws Z_s^* already carry the sampling scale of $\hat{\theta}$.

Proposition 8 (Asymptotic validity)**.** *Let Lemma 1 apply at θ and let Assumption 1 hold. If $\kappa_n \to \infty$ and $\kappa_n / r_n \to 0$, then, with $q_{1-\alpha}(D)$ as in (15), $\mathbb{P}\big[\, \psi_j(\theta) \geq \psi_j(\hat{\theta}) - q_{1-\alpha}(D) \,\big] \to 1-\alpha$ at every θ at which the law of the directional derivative $\psi'_{j,\theta}(G)$, $G \sim N(0, \Sigma)$, is continuous and strictly increasing at its $(1-\alpha)$ quantile.*

Proof. Write $\hat{h}_s = r_n Z_s^*$ and $\varepsilon_n = \kappa_n / r_n$, so that $D_s = r_n^{-1}\big[\psi(\hat{\theta} + \varepsilon_n \hat{h}_s) - \psi(\hat{\theta})\big]/\varepsilon_n$. Conditional on the data $\hat{h}_s \sim N(0, r_n^2 \widehat{\Sigma})$, which converges in probability to the law of G by Assumption 1, so the draws satisfy the bootstrap-consistency requirement for the underlying vector. Since ψ_j is Lipschitz and Hadamard directionally differentiable and $\varepsilon_n \to 0$ with $r_n \varepsilon_n = \kappa_n \to \infty$, the numerical delta method of Hong and Li (2018, Theorem 3.1), which builds on Fang and Santos (2019), gives that the conditional law of $\big[\psi(\hat{\theta} + \varepsilon_n \hat{h}_s) - \psi(\hat{\theta})\big]/\varepsilon_n$ converges in probability to the law of $\psi'_{j,\theta}(G)$. The empirical quantile of $r_n D_s$ therefore converges to $q_{1-\alpha}\big(\psi'_{j,\theta}(G)\big)$ wherever that quantile is a continuity point, and the coverage statement follows from $r_n\big(\psi_j(\hat{\theta}) - \psi_j(\theta)\big) \to_d \psi'_{j,\theta}(G)$, which is the delta method for directionally differentiable maps. □

The coverage statement is pointwise in θ. The finite-sample choice of the step is assessed with the coverage battery used for replication, whose boundary and kink designs probe the configurations where a plain bootstrap fails. The step follows a rule. With asymptotics in the number of strata and r_n of order $\sqrt{H}$, the rule $\kappa_H = H^{1/4}$ satisfies both rate conditions, since $\kappa_H \to \infty$ and $\kappa_H / \sqrt{H} = H^{-1/4} \to 0$. In the implementation $H = 56$, so $\kappa_H = 2.74$, the draws number 4,000, and fixed steps $\kappa \in \{1, 2, 4\}$ are reported as sensitivity, where $\kappa = 1$ is the plain multiplier bootstrap.

The population bound of equation (12) is covered by the same theory. Extending θ with the weighted focal-only total makes the identified share S a smooth ratio of totals, the lower bound $S(1 - \bar{\pi}_{\text{identified}})$ on the population magnitude is that ratio times ψ_j, and Lemma 1 and Proposition 8 apply to the product unchanged.

Applied to the estimation sample at $\kappa_H = 2.74$, the multiplier procedure reproduces the point estimates of Table 3 and delivers the lower bounds reported in the main text, 2.79 percent at the base rung against 2.85 from replication, and 6.45 percent at the finest rung against 7.16, with 2.32 and 5.38 for the population bound against 2.38 and 6.01. The multiplier bounds sit at or slightly below the replication bounds at every rung. In the coverage battery used for replication, which simulates outcomes while holding the weights, the design and the focal-only mass fixed, the minimum coverage over every design and all five rungs is 0.990 at the rule step and 0.990 for the population bound, against a nominal 0.95. The plain multiplier $\kappa = 1$ covers at 0.963 with bounds within two tenths of a percentage point of replication, and moving the step across $\{1, 2, 4\}$ shifts the finest-rung bound by about two thirds of a percentage point in total.

B.5 Coverage validation and stress tests

Table 5 assesses the design-based inference in two steps. Balanced repeated replication agrees with the linearized variance to three decimals on four smooth statistics, which establishes that the replicate construction is sound. Coverage of the one-sided bound for the non-smooth object is then measured by simulating outcomes from designs whose true $1 - \bar{\pi}$ is known by construction while holding fixed the strata, half-samples, weights, the 244 atoms with their belief intervals, and the censoring pattern. Only outcomes are simulated. A target cell bound λ is delivered by choosing the atom's realized rate to invert (7), after which the true $1 - \bar{\pi}$ at each rung follows by aggregation.

The first pass covered the boundary at which every respondent is rationalizable, two interior designs, a design in which a single cell carries the entire shortfall, and a design in which many atoms sit exactly at the non-differentiable point. Minimum coverage was 0.947, taken across all five rungs. At the boundary design the bound never exceeded zero.

The stress test extends this in two directions. Designs are calibrated by bisection so the true shortfall at the finest rung equals 0.050, 0.065 and 0.080, and a further design sets it at the empirical estimate itself. Within-household outcome dependence is induced by a shared-uniform copula, in which member i of household h uses the household draw U_h with probability ρ and an independent draw otherwise, which leaves every atom's marginal probability exactly unchanged while generating positive within-household dependence. The parameter ρ is the mixing probability of that construction, not a Pearson correlation, and the induced outcome correlation depends on the atom's marginal. Each of the nine combinations uses 1,000 Monte Carlo replications. The minimum coverage anywhere in this grid is 0.976 against a nominal 0.95, and 0.982 at the empirical estimate, both above the 0.947 of the first pass, and dependence up to $\rho = 0.7$ does not reduce it. The first-pass minimum is within Monte Carlo error of the nominal level at three hundred replications per design. The procedure is conservative, with the mean bound running well below the truth in interior designs, so the reported figures understate the shortfall. These batteries assess coverage on the designs they contain, which span the configurations the data come close to, and they do not establish coverage uniformly over all distributions.

Table 5: Validation of the design-based inference

Panel A. Replication against linearization, smooth statistics			
Statistic	Estimate	BRR s.e.	Taylor s.e.
Weighted mean survival to target, resolved cases	0.4613	0.00751	0.00750
Weighted share in fair or poor health	0.2277	0.00517	0.00516
Weighted share reporting a belief of one half	0.1819	0.00481	0.00479
Weighted share aged eighty or above	0.0646	0.00274	0.00273
Panel B. Coverage of the nominal 95% *one-sided bound*			
Simulation design	True $1 - \bar{\pi}$ at Z_4	Household ρ	Coverage
Boundary, every respondent rationalizable	0	0	0.997
Interior, small shortfall spread across cells	0.0151	0	0.947
Interior, larger shortfall spread across cells	0.0372	0	0.980
Single binding constraint	0.0200	0	1.000
Many cells at the non-differentiable point	0.0165	0	0.953
Empirical range	0.050 to 0.080	0, 0.3, 0.7	≥ 0.976
Empirical value	0.0828	0, 0.3, 0.7	≥ 0.982

Notes. Panel A compares balanced repeated replication over the 56 strata and their half-samples with the linearized variance for a stratified design with two units per stratum. Agreement on smooth statistics establishes that the replicate construction is sound but says nothing about a non-differentiable functional, which is why Panel B is necessary. Panel B simulates outcomes from designs whose true $1 - \bar{\pi}$ is known by construction while holding fixed the strata, half-samples, weights, the 244 atoms with their belief intervals and the censoring pattern. The first five rows use three hundred replications each. The last two induce within-household outcome dependence through a shared-uniform copula that leaves every atom's marginal probability unchanged, with 1,000 replications for the empirical range, whose three designs reach up to the finest rung's estimate, and five hundred at the empirical value itself. The parameter ρ is the mixing probability of that copula, not a Pearson correlation. Each entry is the lowest coverage across the five rungs. The smallest, 0.947, lies within Monte Carlo error of the nominal 0.95 at three hundred replications. The designs assess coverage where the functional is hardest and do not establish uniform coverage over every data-generating process. The inference procedure was not modified in response to any of these results.

C Belief Measurement

C.1 Focal responses and the grid rule

Subjective probabilities in the Health and Retirement Study are heaped. Among respondents in the estimation sample the three largest exact response combinations are a survival belief of one half, of one, and of zero. A respondent who reports zero and is treated as holding the point belief zero must, under rational expectations, place no mass whatever on the event, so a single occurrence in that cell is unrationalizable. Symmetrically a report of one hundred admits no failures. Because heaping concentrates at exactly these points, the measurement model determines how much of the identified shortfall is attributable to reporting behaviour.

The classification rule is fixed before estimation and uses every subjective-probability question the respondent answered at the anchor wave, on average 6.0 questions. Let A_i denote the set of answers. The reporting grid is

$$g_i = \begin{cases} 1 & \text{if some } x \in A_i \text{ is not a multiple of 5,} \\ 5 & \text{else if some } x \in A_i \text{ is a multiple of 5 but not of 10,} \\ 10 & \text{else if some } x \in A_i \text{ is a multiple of 10 and not in } \{0, 50, 100\}, \\ 50 & \text{otherwise.} \end{cases}$$

The finest granularity a respondent ever displays reveals the grid on which they are answering, so coarseness is inferred from the respondent's own pattern. Table 1 reports the distribution.

A report x_i from a respondent with grid g_i becomes $[\, x_i - g_i/2,\ x_i + g_i/2 \,]$ truncated to $[0, 1]$. This is the interval within which the underlying point belief is taken to lie, and it enters the bound through Proposition 4. A zero from a grid-one respondent stays within half a percentage point of zero. A zero from a grid-ten respondent becomes $[0, 0.05]$. A zero from a focal-only respondent would become $[0, 0.25]$, but focal-only respondents are excluded from the authoritative specification.

These intervals are respondent-specific and inferred from respondent behaviour. Widening a cell to the edges of an analyst-chosen partition, and then permitting the program to select the most favourable belief inside it, would treat the analyst's binning choice as the respondent's own uncertainty, and only the latter is a measurement model.

C.2 Rounding versus imprecision

The grid rule identifies how coarsely a respondent reports. It does not identify whether a focal-only respondent holds a precise point belief and rounds it, or holds an imprecise belief that no point summarizes. Direct probing of probabilities about late-onset dementia finds both types, in roughly equal proportions (Giustinelli et al., 2022). For the second type the benchmark $p = \mathbb{P}(Y \mid I)$ is not well defined, so the point-belief version of rational expectations has no testable content, and counting such respondents as rationalizable by default would inflate $\bar{\pi}$ for a reason unrelated to their beliefs.

The authoritative specification therefore removes focal-only respondents from point-belief inference and reports them separately. Of the 16,780 respondents who report a survival belief and satisfy every other sample criterion, 3,218, or 19.2 percent, report only on the focal grid. Removing them leaves the 13,562 respondents of the estimation sample.

C.3 Precise reporters

If the movement between the literal and interval specifications were an artifact of widening intervals, then restricting to respondents who report on a fine grid, with no widening at all, should not reproduce it. Table 6 shows that it very nearly does. Precise reporters taken literally give a bound of 0.880 against 0.892 when every respondent is given their own interval. Two specifications with opposite mechanics agree, so precise and coarse reporters differ in their beliefs and outcomes, consistent with evidence that, among Finnish men, the share who report round inflation forecasts falls with measured IQ (D'Acunto et al., 2023).

D Data Construction

Beliefs, conditioning variables, weights and design identifiers come from the RAND HRS Longitudinal File, which places the core interviews on harmonized definitions across waves (Health and Retirement Study, 2025; RAND Center for the Study of Aging, 2025). The survival realization also uses the HRS Tracker, the 2024 core interview and the National Death Index linkage the Tracker carries (Health and Retirement Study, 2026b,a).

D.1 Beliefs and realization

The survival belief is the reported percent chance of living to a target age, divided by one hundred. The instrument adapts that target to the respondent's current age and records it, so the event being forecast differs across respondents and the realization has to be constructed individually.

The reporting-grid classification of Appendix C uses every subjective-probability item the anchor wave contains. These are the chances of living to the adapted target age and to age seventy-five, of entering a nursing home within five years, of working after age sixty-two and after age sixty-five, of leaving a bequest of at least ten thousand dollars, of leaving one of at least one hundred thousand dollars, of leaving any bequest at all, and of receiving an inheritance. Responses outside $[0, 100]$ are treated as missing.

Let T_i denote the target age and let A_i denote age at death, which is observed for decedents. Let $\bar{a}_i$ be the oldest age at which the respondent is known to be alive, at an interview between waves 8 and 16, at the last date the Tracker records them alive, or at the 2024 core interview. The

realization is

$$Y_i = \begin{cases} \mathbf{1}\{A_i \geq T_i\} & \text{if } A_i \text{ is observed,} \\ 1 & \text{if } A_i \text{ is missing and } \bar{a}_i \geq T_i, \\ \text{unresolved} & \text{otherwise.} \end{cases}$$

The third case is a respondent alive at last observation who has not yet reached the target age. Such respondents are retained and their mass is allocated as in Proposition 5. They are 30.5 percent of the estimation sample. When the RAND file records no death, A_i can come from the National Death Index, which gives year and quarter of death, and then Y_i is resolved only if the age the quarter allows lies entirely on one side of the target. Appendix H gives the rules and the checks.

D.2 Conditioning variables and sample

The ladder conditions on respondent sex, on age group at the anchor wave, coded as under seventy, seventy to seventy-nine, and eighty or above, on an indicator for fair or poor self-rated health, and on an indicator for current smoking. All four are attributes the respondent unambiguously knows about themselves at the time of the interview, which is the criterion for admitting a variable to the asserted information set.

The estimation sample consists of respondents interviewed at the anchor wave with a non-missing survival belief in $[0, 1]$, a classifiable reporting grid, non-missing sex and age group, and non-missing design identifiers. Respondents who report only on the focal grid are excluded from the authoritative specification and reported separately. This yields 13,562 respondents, of whom 12,745 carry a positive respondent weight.

No respondent is dropped for having an unresolved realization, since that would select on the outcome. Every rung is evaluated on the same 13,562 respondents. Missing sex, age group or design identifiers remove a respondent from the sample entirely, before any rung is computed, and the two health variables are coded as indicators that take the value zero when the underlying item is missing, so that no respondent enters one rung and leaves another. Six respondents have no self-rated health and ninety-five have no smoking status, and they are treated as not in fair or poor health and not currently smoking. Holding the sample fixed allows the decline along the ladder to be read without composition changes, as the nesting in Theorem 1 requires. Respondents with zero or missing survey weight are excluded from weighted quantities by construction.

D.3 Cells

The finest partition consists of 244 atoms defined by the interaction of a coarse belief group with all four conditioning variables. A cell's belief interval is the smallest interval containing its members' own rounding intervals, which contains each member's interval and therefore preserves validity of the bound while avoiding an analyst-chosen band. Coarser rungs are obtained by aggregating atoms and never by re-partitioning, which makes the ladder nested by construction.

E Survey Design and Weighting

E.1 Design identifiers and estimand

The Health and Retirement Study is a stratified multistage sample (Heeringa and Connor, 1995), and the study supplies sampling-error computation codes for design-based variance estimation, a standard-error stratum and, within it, a half-sample (RAND Center for the Study of Aging, 2025). In the estimation sample these give 56 strata with exactly two half-samples each, which is the ultimate-cluster configuration those codes are constructed for. Estimation uses the person-level analysis weight for the anchor wave.

Weighted quantities describe the population represented by positive values of that weight at Wave 7, which is community-dwelling older Americans in 2004, and not the United States population. Two exclusions follow from the weight's construction. Age-ineligible younger spouses carry zero weight, and in this sample they are 817 respondents with mean age 47.7 against 65.5 among the weighted. Institutionalized respondents are represented by a separate nursing-home weight. The difference between the weighted and unweighted columns of Table 3 reflects the first of them.

The largest restriction is that the estimation sample excludes the 3,218 respondents, 19.2 percent of those otherwise eligible, who report only on the focal grid. The weighted estimand is the community-dwelling older population of 2004 among respondents who report on a finer grid. Reweighting cannot recover the full population, because the excluded group is defined by the absence of the object being bounded.

E.2 Replication and linearization

Balanced repeated replication uses a Sylvester Hadamard matrix of order 64 over the 56 strata. In replicate r the half-sample selected by the sign pattern receives twice its analysis weight and the complementary half-sample receives zero, and the variance follows from (13). The number of replicates is the smallest power of two exceeding the number of strata, and the first 56 columns of the matrix are used.

For a stratified design with two units per stratum the linearized variance of a weighted mean reduces to $\sum_h (u_{h1} - u_{h2})^2$, where $u_{hi} = \sum_{j \in hi} w_j (y_j - \hat{\theta}) / \sum_j w_j$ is the unit total of the linearized variable. Panel A of Table 5 compares this with the replication estimator on four smooth statistics. The ratio is 1.00 in every case, which validates the replicate construction. Appendix B assesses coverage for the non-smooth bound by simulation.

E.3 Clustering beyond the household

The exact test assumes independence within a cell. Household dependence is the dominant violation, since 53.0 percent of the estimation sample share a household with another sample member and more than half the cells contain such a pair. Appendix F reports the test with one respondent per household and under a household-clustered bootstrap. Table 2 also reports a design-aware route in

which the same inequalities are evaluated with balanced-repeated-replication standard errors over the strata and half-samples, so that dependence is permitted at the level of the sampling-error half-samples, which nest households. That route replaces the binomial with a normal approximation.

F Additional Robustness

F.1 Where the exact test fails

Figure 6 plots every cell's admissible belief interval against the range of realized rates consistent with its resolved and unresolved outcomes.

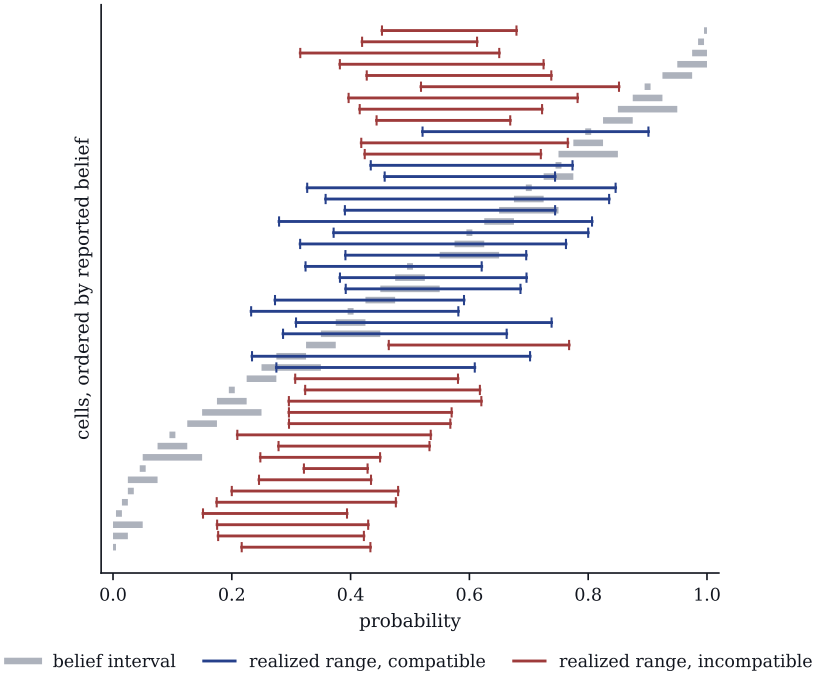


Figure 6: Admissible belief intervals against realized survival ranges, by cell

Notes. Each row is one cell of the authoritative specification, ordered by reported belief. The thick grey segment is the interval of beliefs the cell's respondents could hold given their reporting granularity. The thin segment is the range of realized rates compatible with the cell's resolved outcomes, running from the share known to have reached the target age to that share plus the share whose outcome is still undetermined. A cell is incompatible with rational expectations when the two segments do not overlap. Incompatible cells appear at both ends of the belief distribution. Cells with fewer than twenty-five respondents are omitted.

F.2 Measurement and weighting

Table 6 reports the bound under literal reports, under a restriction to respondents reporting on a grid of five or finer with no interval widening, and under respondent-specific rounding intervals. Literal reports give the tightest bound because a reported zero or one hundred then imposes an absolute restriction. The near-agreement between the precise-reporter and interval specifications,

obtained through opposite mechanics, indicates that the difference from the literal specification reflects heterogeneity between precise and coarse reporters. Inference is reported for the authoritative specification only, because the design-based procedure was assessed for that specification.

The table is computed on the subsample of respondents who report beliefs in both survival and nursing-home entry, so that the three measurement specifications are compared on a fixed sample and the second domain can be shown to behave the same way. Its levels are therefore not comparable with the survival estimates in the main text, which use the full estimation sample.

Table 6: Sensitivity of the bound to the belief-measurement model

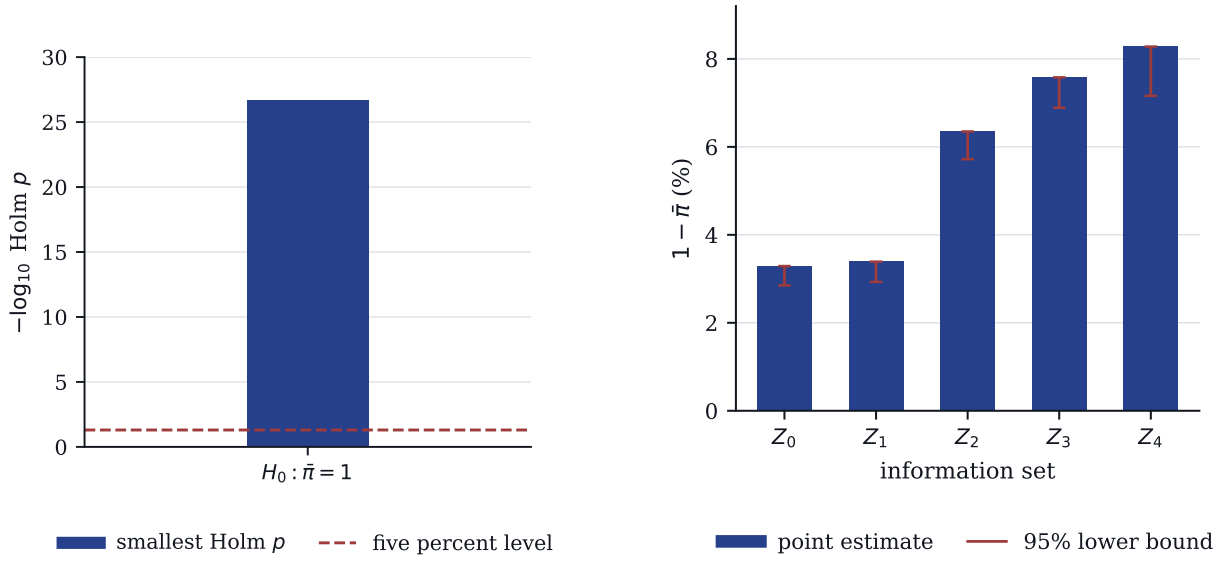
Measurement specification	Survival	Second domain	Status of focal-only reporters
Literal reports	0.826	0.938	retained at face value
Precise reporters only, grid ≤ 5	0.880	0.969	excluded
Respondent-specific intervals	0.892	0.982	retained with own interval

Notes. Point estimates of $\bar{\pi}$, unweighted, computed on the subsample holding beliefs in both domains so that the three measurement specifications in the rows are comparable to one another. The second domain is nursing-home entry within five years and is retained here only to show that the measurement model, not the domain, drives the sensitivity. Inference is not reported for these specifications because the design-based procedure was validated for the authoritative specification only, and reporting a bound without a validated confidence statement would invite the two to be read together. Taking reports literally imposes an absolute restriction on respondents who answer zero or one hundred and therefore yields the tightest bound. Restricting to respondents who report on a grid finer than multiples of ten, with no interval widening at all, gives almost the same answer as giving every respondent their own rounding interval, which indicates that the movement reflects a genuine difference between precise and coarse reporters and not the mechanical effect of widening intervals.

The zero-weight group behind the difference between the weighted and unweighted ladders is described in Appendix E. The ordering across rungs is the same in the two columns, with age producing the largest single change and health and smoking adding further. Sex moves both columns little. The weighted bound falls from 0.9671 to 0.9661, a tenth of a percentage point, and the unweighted bound from 0.9503 to 0.9464, about four tenths.

F.3 Household dependence

Table 2 reports the exact test under four treatments. Retaining one respondent per household, selected by a deterministic identifier rule that does not use the outcome, leaves the smallest Holm-adjusted p-value at 8.4×10^{-58}, which is smaller than the full-sample value, so the rejection is not an artifact of counting spouses twice. A bootstrap that resamples households leaves it at 2.1×10^{-27}, and the half-sample replication route, which permits dependence at the level of the sampling-error half-samples, leaves it at 2.4×10^{-34}. Household and design dependence cost orders of magnitude and leave the conclusion unchanged. Figure 7 places that rejection beside the identified magnitude.



(a) Existence

(b) Magnitude

Figure 7: Strength of the rejection against the identified magnitude

Notes. Panel (a) reports the negative base-ten logarithm of the smallest Holm-adjusted p-value for the hypothesis $\bar{\pi} = 1$ under the household-clustered treatment, the most conservative of the three routes in Table 2. The dashed reference line is the five percent level. Panel (b) reports the weighted point estimate of $1 - \bar{\pi}$ in percent at each rung of the ladder, with whiskers running down to the one-sided 95 percent lower confidence bound from balanced repeated replication. The two panels are on unrelated scales.

F.4 Finite-sample benchmark

Refining cells lowers a plug-in bound mechanically, because $\bar{\lambda}$ is concave in the cell mean and sampling noise therefore biases the plug-in downward, with the bias growing as cells shrink. Under literal beliefs the effect is large. Take each reported belief as an exact point, key cells on the exact report and the conditioning variables, which gives 529 atoms in the estimation sample, and let each respondent's own report generate their outcome with the censoring pattern held fixed. This population is rational by construction, yet its minimum non-rationalizable share rises from 0.21 percentage points at Z_0 to 5.10 at Z_4. The same literal ladder computed on the observed outcomes rises from 6.05 to 11.98 percent, so most of its rise along the ladder would also appear in a population that is rational by construction, and a literal ladder cannot be read as evidence about information sets.

Under the authoritative specification the belief intervals convert the kink of Proposition 2 into the plateau of Proposition 4. On a plateau the bound is locally constant in the cell mean, so the first-order bias vanishes, although a cell mean can still leave the plateau when the cell is small. The same construction on the authoritative cells gives a benchmark of 0.00 percentage points at Z_0 and Z_1, 0.32 at Z_2, 0.76 at Z_3 and 1.34 at Z_4. At the finest rung the benchmark is about a quarter of the literal-belief one and roughly a sixth of the estimate.

That benchmark is also the harsher of two nulls. The coverage designs of Table 5 place each

cell's success probability at the midpoint of its belief interval, where the plateau is widest and the bound is least sensitive to sampling noise. Generating outcomes from each respondent's own belief puts the cell mean wherever the members' beliefs put it, sometimes near an interval edge, so the own-belief null is the less favourable of the two.

F.5 The exact lower bound

The exact route of Appendix B, which places simultaneous Clopper–Pearson intervals on every cell proportion and weight, yields a lower bound on $1 - \bar{\pi}$ of 0.40 percent at the base rung, against an unweighted point value of 6.39 percent on the same cells. It requires no asymptotics and holds under independent sampling for the unweighted empirical distribution, and its conservatism comes from the multiplicity correction.

F.6 Ordering, sharpness and concentration

Table 7 places the authoritative ladder beside the implementation in which the respondent's belief interval enters the partition key. The two differ only in how cells are formed. The authoritative partition groups respondents by a coarse belief group and assigns the cell the smallest interval containing theirs, which lets the cell maximisation choose a belief no member admits. The sharp partition keys on the interval itself, so every cell is homogeneous in $[a, b]$ and the union step disappears. Coarser rungs still aggregate the same atoms in both, so each ladder is nested by construction and monotonicity holds exactly.

Removing the union relaxation can only tighten the population bound, so the sharp column would be the object to report if it could be estimated. The cells that make it sharp are too small to support the plug-in estimator.

Table 7: The ladder under the conservative and the sharp implementation

Rung	Information set Z_j	Conservative	Sharp
Z_0	belief only	3.29	4.45
Z_1	+ own sex	3.39	4.93
Z_2	+ own age group	6.35	8.50
Z_3	+ own self-rated health	7.58	10.02
Z_4	+ own current smoking	8.28	11.51

Notes. Weighted minimum non-rationalizable share $1 - \bar{\pi}$ in percent. The conservative column is the authoritative specification of Table 3, built on 244 atoms whose belief interval is the smallest interval containing their members' intervals. The sharp column keys the partition on the respondent's own interval and uses 889 atoms, which removes the union relaxation. The sharp column does not support reliable inference. Its atoms hold a median of 4 respondents against 20 in the conservative partition, with 67.5 percent holding fewer than ten, and at that cell size the plug-in bound is badly biased. Under a rational population its benchmark reaches 5.75 percentage points at the finest rung, about half of the 11.51 reported, and the one-sided lower bound covers in 0.21 of replications in the design battery and 0.08 in the stress grid, against a nominal 0.95. The conservative partition covers at 0.947 and above throughout.

The increments in Table 3 are order-specific, since each rung conditions on the previous one.

Averaging a variable's marginal contribution over all orderings of the four variables gives the Shapley decomposition in Figure 8, whose components sum by construction to the total rise of 5.00 percentage points.

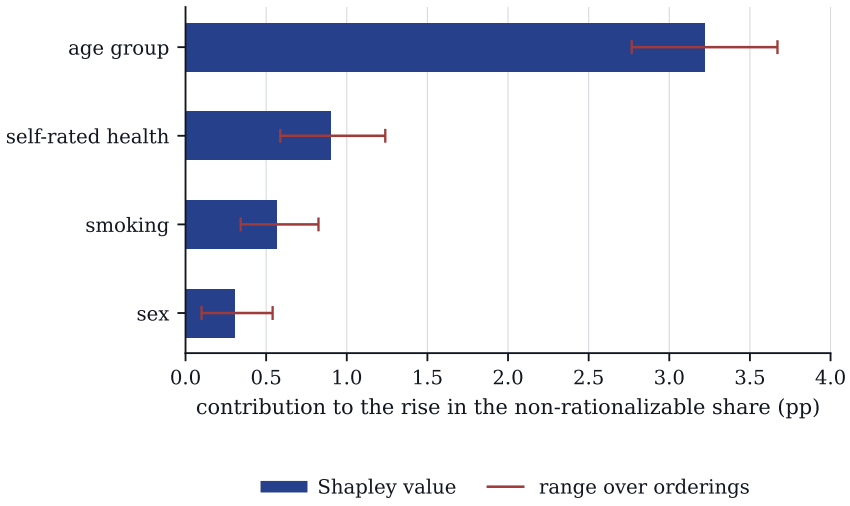


Figure 8: Contribution of each conditioning variable across all orderings

Notes. Bars are Shapley values, the average marginal contribution of each variable to the rise in the weighted minimum non-rationalizable share over the twenty-four orderings of the four conditioning variables, in percentage points. Whiskers run from the smallest to the largest marginal contribution the variable attains at any position. The four bars sum to the total rise of 5.00 points. Every subset bound is computed on the same sample and the same finest partition.

The Shapley values are 3.22 points for age group, 0.90 for self-rated health, 0.57 for smoking and 0.31 for sex. Age group dominates at any position, contributing between 2.77 and 3.67 points, so the ranking in the main text does not depend on the order of the ladder. Every whisker has positive width, so every contribution moves with position. Sex contributes least at every position, between 0.10 points when it enters first and 0.54 when it enters last.

Figure 9 shows where the shortfall comes from. Each cell contributes $w_c(1-\bar{\lambda}_c)$ to the total, and ordering cells from the largest contribution downward traces the cumulative share of the shortfall against the cumulative share of the weighted population.

Without conditioning, the shortfall is the work of a few cells at the tails of the belief distribution, the largest carrying 36.0 percent of it, and dropping the five largest contributors leaves 0.00 percentage points. With the full information set, 173 of 244 cells contribute, the largest carries 6.1 percent, and dropping the ten largest still leaves 5.95 of 8.28 points.

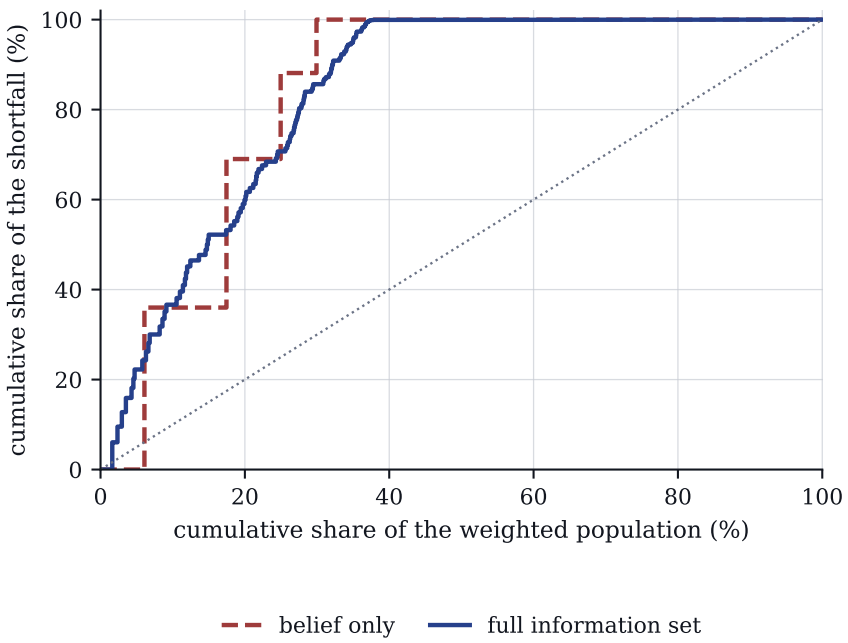


Figure 9: Concentration of the shortfall across cells

Notes. Cells are ordered by their contribution $w_c(1 - \bar{\lambda}_c)$ to the weighted minimum nonrationalizable share, largest first. The horizontal axis is the cumulative share of the weighted population, the vertical axis the cumulative share of the total shortfall. The dotted line marks proportionality, where a given share of the population accounts for the same share of the shortfall. Both curves flatten near a third of the population, since only cells with $\bar{\lambda}_c < 1$ contribute at all. They differ in the size of the steps. The base rung curve climbs in 4 jumps, the first of which is 36.0 percent of the total, while the finest rung curve climbs in 173 small ones.

F.7 Where the shortfall sits

Each cell of the finest partition belongs to exactly one category of each conditioning variable, so summing $w_c(1-\bar{\lambda}_c)$ within a category decomposes the total exactly, once per dimension, into each group's contribution to the population bound.

Age organizes most of the shortfall. Respondents aged seventy to seventy-nine are 17.0 percent of the weighted population and account for 48.8 percent of the shortfall. Those eighty or older are 6.5 percent and account for 25.8. Together, under a quarter of the population carries about three quarters of it, while the 76.6 percent under seventy carry 25.4 percent. Fair or poor health is mildly disproportionate, 22.8 percent of the population against 37.0 of the shortfall. Smoking and sex are close to proportional. Across belief bands the contributions track population shares except at the top, where the 6.1 percent reporting a survival chance above ninety-five carry 14.3 percent.

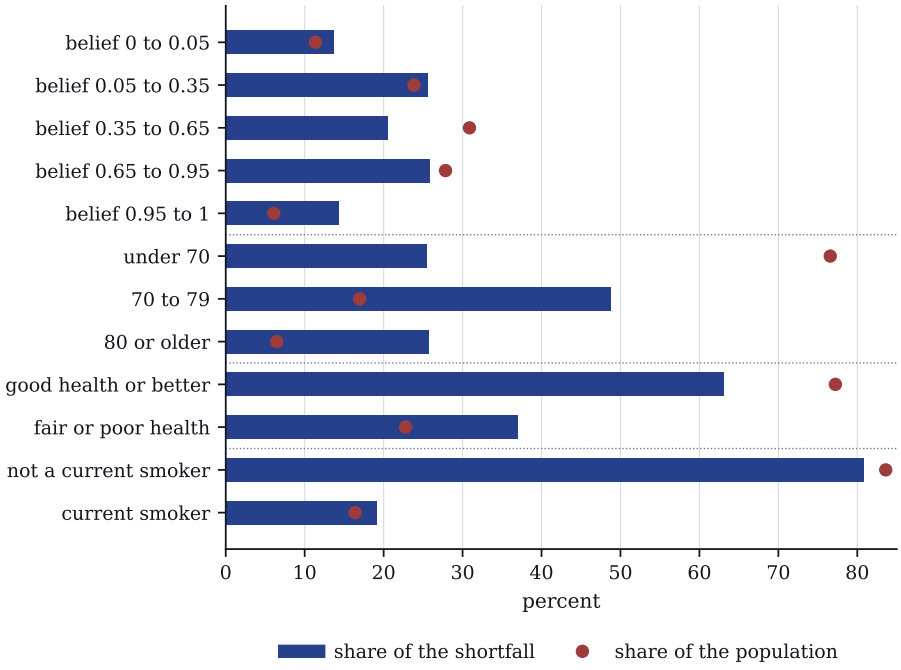


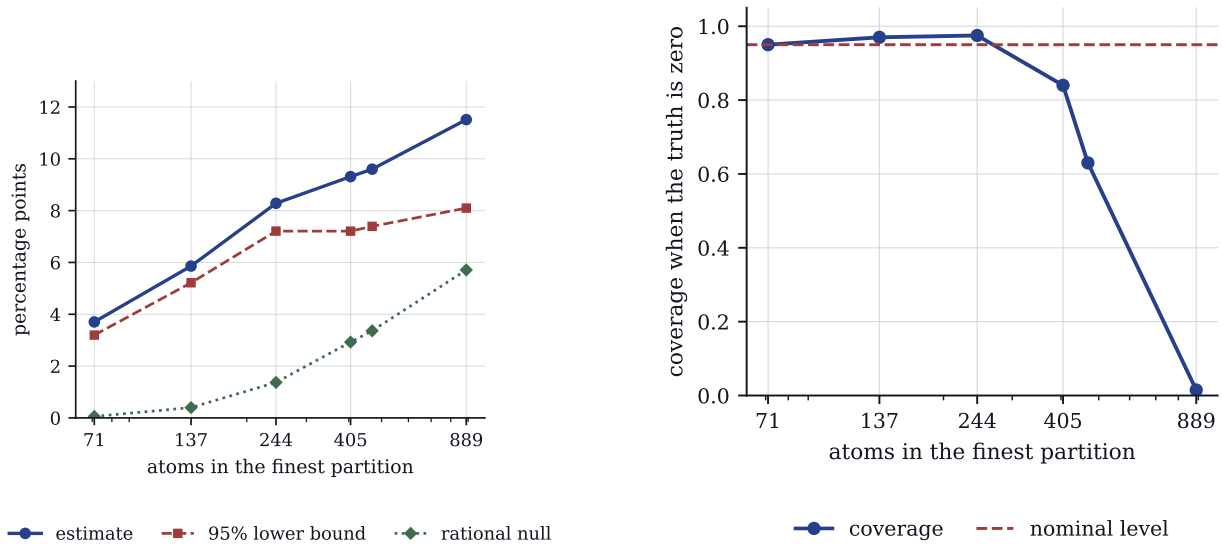
Figure 10: Contribution to the shortfall by group

Notes. Bars are each group's share of the total weighted shortfall at the finest rung, computed as the sum of $w_c(1-\bar{\lambda}_c)$ over the cells belonging to that group. Circles are the group's share of the weighted population. A bar exceeding its circle means the group contributes more than its size. Dotted lines separate dimensions, and within each dimension the bars sum to one hundred. Sex is omitted from the figure because both categories are within three points of proportional. The rational-null benchmark is below 0.8 percentage points for every group, so it does not reorder them.

F.8 The identification-precision frontier

The conservative and sharp partitions of Table 7 are two points on a curve. Grouping the reported belief to the nearest half, fifth, tenth, twentieth or fiftieth, and finally keying on the respondent's

own interval, gives a family of partitions running from 71 to 889 atoms. Figure 11 traces the estimate, its lower bound, the rational-null benchmark and the coverage of the one-sided bound when the truth is zero, which is the case where a false positive would be most damaging.



(a) Identification

(b) Precision

Figure 11: Refining the partition tightens the bound and weakens the inference

Notes. Each point is one partition of the belief, from the nearest half to the respondent's own interval, always interacted with the same four conditioning variables and evaluated at the finest information set. Panel (a) reports the weighted estimate of $1 - \bar{\pi}$, its one-sided 95 percent lower bound, and the shortfall a population that is rational by construction produces on the same cells. Panel (b) reports the share of simulations in which the lower bound stays below the truth when every respondent is rational, against the nominal level. The horizontal axis is on a log scale.

The estimate rises monotonically with refinement, from 3.70 percent at 71 atoms to 11.51 at 889, because each refinement removes some of the union relaxation. The rational-null benchmark rises with it, and faster in proportion. Coverage is 0.950 at the coarsest partition, reaches 0.975 at 244 atoms, and then falls away, to 0.840 at 405, 0.630 at 469 and 0.015 at the respondent's own interval. Coverage, and not the estimate, limits how fine the partition can be.

The partition used in the main text, beliefs grouped to the nearest tenth, was fixed in advance and is the finest that retains nominal coverage at the boundary. Coverage under a rational null requires no knowledge of the true shortfall, so the curve measures the reliability of the inference independently of the estimate.

F.9 What the conventional regression sees

The classical way to test efficiency is to regress the outcome on the reported belief and the asserted information,

$$Y_i = \alpha + \beta\, p_i + \gamma' Z_i + \varepsilon_i, \tag{16}$$

and to read rational expectations as $\alpha = 0$, $\beta = 1$ and $\gamma = 0$. This treats the report as an exact point belief, so the exercise is a literal-report diagnostic, and it has to respect censoring.

The coefficient is linear in the outcome vector,

$$\hat{\beta}_j \;=\; \sum_i a_{ji}\, Y_i, \qquad a_j \;=\; \text{the } j\text{th row of } (X'WX)^{-1}X'W,$$

and the weights attached to unresolved observations differ in sign. Setting every unresolved outcome to zero and then to one therefore does not bracket the coefficient. It evaluates two arbitrary points inside the feasible set. The sharp interval assigns each unresolved outcome by the sign of its own weight, one to the positive weights and zero to the negative ones to maximize and the reverse to minimize. Since the free outcomes are binary, enter linearly and are unconstrained, those assignments attain the extrema exactly.

Table 8: Forecast-efficiency coefficients, sharp bounds under censoring

	Survival to target (27.1% unresolved)	Work after 62 (9.3% unresolved)
Reported belief p	$[-0.430,\ 0.698]$	$[0.295,\ 0.522]$
Female	$[-0.345,\ 0.442]$	$[-0.140,\ 0.058]$
Second age group	$[-0.270,\ 0.257]$	$[-0.092,\ 0.097]$
Third age group	$[-0.572,\ -0.034]$	$[-0.074,\ 0.091]$
Fair or poor health	$[-0.492,\ 0.178]$	$[-0.235,\ -0.042]$
Current smoker	$[-0.571,\ 0.268]$	$[-0.145,\ 0.052]$

Notes. Weighted least squares of equation (16). Each entry is the sharp interval over all assignments of the unresolved outcomes, obtained by assigning each one according to the sign of its influence weight on that coefficient. Age groups are the second and third categories of each sample's own grouping. Rational expectations relative to the asserted information set implies a coefficient of one on the reported belief and zero on everything else.

The conventional test identifies almost nothing here. In the survival domain, where over a quarter of outcomes are unresolved, only the oldest age group has an interval that excludes zero, and the interval on the reported belief runs from -0.430 to 0.698, which contains both zero and one, so the regression can neither reject nor establish $\beta = 1$. In the work domain, where under a tenth is unresolved, two intervals exclude zero. The coefficient on the reported belief lies in $[0.295, 0.522]$, which excludes both zero and one, and fair or poor health lies in $[-0.235, -0.042]$, which excludes zero. Sex and smoking do not.

On the same survival data the bound framework delivers 3.29 percent at the base rung and 8.28 at the finest, with lower bounds that have been checked for coverage, while the efficiency regression signs one coefficient and leaves the reported belief undetermined. The regression needs a conditional mean, which unresolved outcomes leave unidentified. The bound needs only whether the reachable range of outcomes meets the admissible belief interval, and unresolved outcomes widen that range.

G Joint Rationalizability Across Domains

The framework extends to several outcomes elicited from the same individuals. A large share of a population may be rationalizable in each domain separately while a much smaller share is rationalizable in all of them at once, and the single-domain bound cannot detect the difference.

G.1 The joint program

With D binary domains, let $y \in \{0,1\}^D$ index outcome configurations and let g_y denote the observed mass on configuration y within a cell defined by the vector of reported beliefs $(p_1, \ldots, p_D)$ and covariates. Write r_y for the mass of the rational subpopulation at configuration y. Working directly in these masses, instead of in a share times a conditional law, keeps the problem linear:

$$\max_{r \geq 0} \sum_y r_y \quad \text{subject to} \quad r_y \leq g_y \;\; \forall y, \qquad \sum_y y_d \, r_y \;=\; p_d \sum_y r_y \;\; \forall d. \tag{17}$$

Writing the program as a share λ multiplying a free joint law q with marginals p_d would make the marginal constraint bilinear in (λ, q), and the substitution $r = \lambda q$ removes the product. Every coefficient in (17) is then either an indicator or a reported belief, so the program has known coefficients and an estimated right-hand side.

Three extensions preserve linearity. Interval beliefs replace the equality by $a_d \sum_y r_y \leq \sum_y y_d r_y \leq b_d \sum_y r_y$. Censoring enters exactly as in Proposition 5, through allocation variables $\alpha_{g,y} \geq 0$ for each group g of respondents with a partially known configuration, with $\sum_y \alpha_{g,y}$ equal to that group's mass, $\alpha_{g,y} = 0$ for configurations incompatible with what is known, and the componentwise constraint replaced by $r_y \leq \sum_g \alpha_{g,y}$. Weights enter through the right-hand side.

A natural benchmark asks how large the jointly rational share could be if the analyst preserved each domain's observed marginal outcome distribution but were free to choose the cross-domain dependence.

Definition 2 (Marginal envelope)**.** $\pi_{\text{marg}} \;=\; \mathbb{E}\Big[\min_d \bar{\lambda}_d(X)\Big]$, where $\bar{\lambda}_d$ is the single-domain cell bound of Section 3.

Proposition 9 (Envelope)**.** *π_{marg} is exactly the largest jointly rational share attainable while preserving each domain's marginal outcome distribution and choosing the coupling freely. Consequently*

$$\pi_{\text{joint}} \;\leq\; \pi_{\text{marg}} \;\leq\; \min_d \bar{\pi}_d. \tag{18}$$

The proof is at the end of this appendix. The decomposition

$$\underbrace{1 - \pi_{\text{marg}}}_{\text{marginal inconsistency}} \qquad \text{and} \qquad \underbrace{\pi_{\text{marg}} - \pi_{\text{joint}}}_{\text{additional joint restriction}} \tag{19}$$

separates what the marginals already imply from what the observed dependence adds.

G.2 Two domains

For $D = 2$ the program admits a closed form, so no solver is required.

Proposition 10 (Two-domain characterization)**.** *Let $\delta = p_1 + p_2 - 1$ and let $(g_{00}, g_{01}, g_{10}, g_{11})$ be the observed cell masses. Of the sixteen pairwise feasibility conditions implied by (17), twelve are vacuous or reproduce the four inequalities defining $\bar{\lambda}_1$ and $\bar{\lambda}_2$. The remaining four are*

$$S \le \frac{g_{11}}{p_1 + p_2 - 1} \qquad \text{if } p_1 + p_2 > 1, \tag{20}$$

$$S \le \frac{g_{00}}{1 - p_1 - p_2} \qquad \text{if } p_1 + p_2 < 1, \tag{21}$$

$$S \le \frac{g_{10}}{p_1 - p_2} \qquad \text{if } p_1 > p_2, \tag{22}$$

$$S \le \frac{g_{01}}{p_2 - p_1} \qquad \text{if } p_2 > p_1, \tag{23}$$

and the cell's jointly rational mass is

$$S^*(x) \;=\; \min\Big\{\bar{\lambda}_1,\ \bar{\lambda}_2,\ 1,\ \textit{whichever of (20) to (23) apply}\Big\}.$$

Writing $S^{\mathrm{marg}}(x) = \min\{\bar{\lambda}_1, \bar{\lambda}_2, 1\}$ for the cell-level envelope, $S^*(x) = S^{\mathrm{marg}}(x)$ if and only if every applicable constraint among (20) to (23) is slack at $S^{\mathrm{marg}}(x)$. The population objects of Definition 2 and Proposition 9 are the expectations of these cell quantities over the distribution of cells, so $\pi_{\mathrm{joint}} = \mathbb{E}[S^*(X)]$ and $\pi_{\mathrm{marg}} = \mathbb{E}[S^{\mathrm{marg}}(X)]$.

The reduction was verified numerically against a linear-programming solution on four hundred randomly generated full-support cells with random beliefs, with a maximum absolute discrepancy of order 10^{-16}.

The four constraints are Fréchet restrictions. A rational subpopulation whose two marginals are p_1 and p_2 must place at least $S(p_1 + p_2 - 1)$ of its mass on joint occurrence, at least $S(1 - p_1 - p_2)$ on joint non-occurrence, and at least $S|p_1 - p_2|$ on the corresponding discordant configuration. The data must supply that mass, and when they do not the joint bound falls well below the envelope.

Example 1. Let $p_1 = p_2 = 0.9$ and $(g_{00}, g_{01}, g_{10}, g_{11}) = (0.11, 0.44, 0.44, 0.01)$, so every configuration has positive mass. Both marginals equal 0.45, hence $\bar{\lambda}_1 = \bar{\lambda}_2 = 0.5$ and $\pi_{\mathrm{marg}} = 0.500$. But (20) requires $r_{11} \ge 0.8\,S$ while only $g_{11} = 0.01$ is available, so $\pi_{\mathrm{joint}} \le 0.0125$. The joint restriction reduces the bound by a factor of forty on data with full support.

Example 1 shows the joint object is not a relabelling of the minimum of the marginal bounds. Two mechanisms activate the extra constraints. Beliefs far from one half activate the co-occurrence and non-occurrence bounds (20) and (21), since the required joint mass $S(p_1 + p_2 - 1)$ is positive only when $p_1 + p_2 > 1$ and $S(1 - p_1 - p_2)$ only when $p_1 + p_2 < 1$. Beliefs that differ across domains activate the discordance bounds (22) and (23), which require mass $S|p_1 - p_2|$ on the discordant configuration and apply even when both beliefs sit near one half, as at $p_1 = 0.45$ and $p_2 = 0.55$.

Whether an applicable restriction actually binds in a given cell is a property of that cell's observed masses, as Proposition 10 states, and the population gap $\pi_{\text{marg}} - \pi_{\text{joint}}$ averages those cell-level shortfalls. The beliefs determine which restrictions apply, and none applies when $p_1 + p_2 = 1$ with $p_1 = p_2$, so the joint object is weakest when beliefs are near one half *and* close to one another. For D domains the analogous condition for the all-ones constraint is that the mean belief exceed $(D-1)/D$, which is 0.50, 0.67 and 0.75 at two, three and four domains, so the belief levels required for the joint object to apply rise as domains are added. Domains whose beliefs cluster near one half and resemble each other reproduce the marginal envelope at any sample size.

In simulated populations in which a known share is rational and the non-rational share fails in all domains at once or independently across them, the joint bound stays close to the marginal envelope when beliefs are drawn from a low range and separates from it as the range rises. With three domains, beliefs in the upper range and independent failures, the joint bound returns roughly 0.23 where the marginal envelope returns roughly 0.50 and the true jointly rational share is 0.125, so the bound covers most of the distance to the truth.

G.3 Empirical application

Proposition 10 also guides the choice of domains. The joint bound is informative for domains whose beliefs are dispersed and whose levels differ, so that the co-occurrence and discordance requirements apply to a large share of the population. That criterion involves only the belief distributions, and the pair used here was selected by applying it to those distributions before any joint bound was computed. The study elicits a second personal probability that meets the criterion, the chance of working full time after age sixty-two, asked of respondents who have not yet reached that age.

That belief has a mean near one half and a standard deviation of 0.36, against survival beliefs concentrated well above one half among the same respondents, so the two levels differ in the direction the criterion asks for. Among the 3,397 respondents who report both on a non-focal grid, 49.1 percent of the weighted mass sits in cells where the two beliefs sum to more than one, which is where (20) applies.

The additional restriction is small. Over 55 cells the maximum rationalizable share is 0.9951 for survival and 0.8975 for work, the marginal envelope is 0.8972, and the joint bound is 0.8938. Requiring the same people to be rationalizable in both domains at once therefore costs 0.34 percentage points beyond what the two marginals already imply. A population that is rational by construction returns 0.00 percentage points for the same object on the same cells, so the gap is not a finite-sample artifact, and it is an order of magnitude below the marginal inconsistency it is added to.

The constraints apply widely here, and they bind only where the data fail to supply the required joint mass. In this population the data mostly do supply it, because respondents who report a high survival chance together with a low chance of working are common enough to fill the discordant configurations the Fréchet inequalities demand. The joint object is active and identified and still close to the envelope.

Nursing-home entry is the contrast case, a pair that meets the criterion in almost no cell. Applied to survival and nursing-home entry, the joint bound falls below the marginal envelope by under two percentage points, and nearly the entire gap comes from the single cell in which the two beliefs sum to more than one. Survival beliefs in this population are dispersed but nursing-home beliefs are concentrated near zero, so the co-occurrence restriction is inactive almost everywhere.

Selecting domains on the position of their belief distributions moves the share of the population facing an applicable constraint from almost nothing to about half, and selecting them on availability alone does not. In neither pair does the applicable constraint bind hard, while Example 1 shows the same restriction cutting a bound by a factor of forty on data with full support.

G.4 Proofs

Proof of Proposition 9

Upper direction. Let r solve (17) with total mass S. For each domain d, summing the componentwise constraints over configurations with $y_d = 1$ gives $p_d S = \sum_y y_d r_y \le \sum_{y:y_d=1} g_y = m_d$, and over configurations with $y_d = 0$ gives $(1-p_d)S \le 1 - m_d$. Hence $S \le \bar{\lambda}_d$ for every d and therefore $S \le \min_d \bar{\lambda}_d$.

Attainability. Fix the cell's marginals m_d and set $S = \min_d \bar{\lambda}_d$. Let Q be any joint law on $\{0,1\}^D$ with marginals p_d, which exists, for instance the product law. Let A be any non-negative measure of total mass $1 - S$ with marginals $m_d - Sp_d$, which exists because those quantities are non-negative exactly when $S \le m_d/p_d$ and their complements $1 - m_d - S(1-p_d)$ are non-negative exactly when $S \le (1-m_d)/(1-p_d)$, both implied by $S \le \bar{\lambda}_d$. Then $G = S\,Q + A$ has marginals m_d and admits rational mass S. Since the analyst is free to choose the coupling, π_{marg} is attained. The chain (18) follows because $\mathbb{E}[\min_d \bar{\lambda}_d] \le \mathbb{E}[\bar{\lambda}_d] = \bar{\pi}_d$ for each d. □

Proof of Proposition 10

Write $t = r_{11}$. The marginal constraints determine the remaining masses, $r_{10} = Sp_1 - t$, $r_{01} = Sp_2 - t$ and $r_{00} = S(1 - p_1 - p_2) + t$. Non-negativity of the four masses gives

$$t \ge 0, \quad t \le Sp_1, \quad t \le Sp_2, \quad t \ge S\delta,$$

and the componentwise constraints $r_y \le g_y$ give

$$t \le g_{11}, \quad t \ge Sp_1 - g_{10}, \quad t \ge Sp_2 - g_{01}, \quad t \le g_{00} + S\delta.$$

Feasibility of S is therefore equivalent to

$$\max\{0,\ S\delta,\ Sp_1 - g_{10},\ Sp_2 - g_{01}\} \ \le\ \min\{Sp_1,\ Sp_2,\ g_{11},\ g_{00} + S\delta\},$$

which is a system of sixteen pairwise inequalities. Four are vacuous, namely $0 \le Sp_1$, $0 \le Sp_2$, $0 \le g_{11}$ and $S\delta \le g_{00} + S\delta$. Two hold because $p_d \le 1$, namely $S\delta \le Sp_1$ and $S\delta \le Sp_2$. Two are trivial identities, $Sp_1 - g_{10} \le Sp_1$ and $Sp_2 - g_{01} \le Sp_2$. Four reproduce the single-domain constraints, since $Sp_1 - g_{10} \le g_{11}$ is $S \le m_1/p_1$, $Sp_2 - g_{01} \le g_{11}$ is $S \le m_2/p_2$, $Sp_1 - g_{10} \le g_{00} + S\delta$ is $S \le (1-m_2)/(1-p_2)$, and $Sp_2 - g_{01} \le g_{00} + S\delta$ is $S \le (1-m_1)/(1-p_1)$. The remaining four are (20) to (23), each active only under the stated sign condition. Hence the cell value is the minimum of $\bar{\lambda}_1$, $\bar{\lambda}_2$, one and the applicable members of (20) to (23), and equality with the envelope holds exactly when all applicable members are slack at $S = \min\{\bar{\lambda}_1, \bar{\lambda}_2, 1\}$. □

Verification of Example 1

With $p_1 = p_2 = 0.9$ and $(g_{00}, g_{01}, g_{10}, g_{11}) = (0.11, 0.44, 0.44, 0.01)$ the marginals are $m_1 = g_{10} + g_{11} = 0.45$ and $m_2 = g_{01} + g_{11} = 0.45$, so $\bar{\lambda}_1 = \bar{\lambda}_2 = \min\{0.45/0.9,\ 0.55/0.1\} = 0.5$ and $\pi_{\text{marg}} = 0.500$. Here $\delta = 0.8 > 0$, so (20) applies and gives $S \le g_{11}/\delta = 0.01/0.8 = 0.0125$. All four configurations carry positive mass, so the gap is not a consequence of degenerate support. □

H Survival Ascertainment

The survival realization combines the RAND file with the HRS Tracker, the 2024 core interview and the National Death Index. The 2004 beliefs, belief intervals, cells, weights and design identifiers are those of Appendix D throughout, and only the realization changes.

H.1 Rules

Ages are in completed years, as in the RAND file. An outcome the RAND file records is never changed. An outcome the RAND file leaves unresolved is resolved by the following rules.

Alive evidence. The Tracker records the last date on which the respondent was known to be alive, taken from the respondent's own core interview, a spouse's or partner's interview reporting them alive, or a field contact with someone who knows them directly (Health and Retirement Study, 2026b). The 2024 core interview records the respondent's age (Health and Retirement Study, 2026a). The age at the last-alive date is computed from the Tracker's birth year and month, and when either month is missing the lower of the two possible ages is used. The outcome is resolved as reached when the larger of these ages is at least the target.

National Death Index. The Tracker carries the year and calendar quarter of death for respondents matched to the National Death Index, matches that the National Center for Health Statistics flags as valid and HRS staff verify. Age at death is bracketed by the year difference and narrowed by the quarter. When the birth month falls before the death quarter, the birthday of that year has passed and the age is exact. When it falls after the quarter, the birthday has not yet occurred and the age is again exact. When it falls inside the quarter, the age is one of two adjacent values. The

outcome is resolved as not reached when the upper value is below the target, as reached when the lower value is at least the target, and left unresolved otherwise.

Conflicts. A National Death Index death in a year earlier than alive evidence from another source leaves the outcome unresolved, since one of the two sources must be wrong and the data do not say which.

H.2 What each source resolves

Of the 13,562 respondents in the authoritative sample, the RAND file leaves 38.1 percent unresolved. Alive evidence resolves 659 of them as having reached the target, 406 through the 2024 interview, 103 through Tracker contacts in 2023 and 5 through contacts in 2025, with the remainder dated earlier. The National Death Index resolves 368 more, 237 as deaths before the target and 131 as deaths at or after it. In total 1,027 outcomes are resolved, and the unresolved share falls to 30.5 percent of the sample and 39.4 percent of its weight. Table 9 reports the resulting shares and bounds under each combination of sources. Adding National Death Index deaths to the RAND file gives 2.64 percent at the base rung and 7.55 at the finest, adding only the alive evidence gives 2.96 and 7.43, and every source together gives 3.29 and 8.28.

Table 9: Survival outcomes and the bound under each source of follow-up

	Known outcome, weighted		Unresolved		$1-\bar{\pi}$ (%)	
Outcome coding	reached	did not	weighted	unweighted	Z_0	Z_4
RAND longitudinal file	22.0	30.6	47.4	38.1	2.24	6.50
RAND and alive evidence	27.2	30.6	42.2	33.2	2.96	7.43
RAND and National Death Index	23.2	32.7	44.2	34.9	2.64	7.55
All sources	28.0	32.7	39.4	30.5	3.29	8.28

Notes. Shares in percent. Weighted shares use the respondent weight over the 12,745 respondents with a positive weight, and unweighted shares are over the 13,562 respondents of the authoritative sample. Alive evidence is the last date the HRS Tracker records the respondent alive and the age at the 2024 core interview. The National Death Index supplies year and quarter of death. Each source only resolves outcomes the RAND file leaves open, and the last two columns are point values of the minimum non-rationalizable share at the base and finest rungs. Appendix H gives the rules, the conflicts between sources and their resolution.

H.3 Disagreements between sources

The RAND file and the National Death Index. On 13 respondents the two sources imply opposite outcomes. Birth year and month agree between the RAND file and the Tracker in every one of them, so the disagreement is about the date of death and not about identity. In ten of them the two death years differ by at most two years, so the disagreement concerns which side of the birthday on which the target is reached the death fell. The RAND value is kept because it comes from exit interviews, which report the date of death directly. Leaving all 13 outcomes unresolved

moves the minimum non-rationalizable share to 3.26 percent at the base rung and 8.25 at the finest, against 3.29 and 8.28.

The National Death Index and alive evidence. 18 respondents have a National Death Index death in a year before a date at which another source records them alive. Every such alive record comes from a field contact. These outcomes stay unresolved.

Ambiguous ages. After the quarter is used, 6 deaths remain whose age at death could fall on either side of the target. They stay unresolved.

Match quality. The median National Death Index match score among the resolved deaths is 79, and 50.8 percent of them score below eighty. Discarding the 180 matches with a score below eighty among respondents with a positive weight gives 3.10 and 7.71 percent, still above the 2.24 and 6.50 of the RAND file alone.

H.4 Retrospective follow-up

Figure 4 recomputes the outcome using only evidence dated by each year: RAND deaths with a death year no later than the cutoff, RAND interviews ending by it, National Death Index deaths in years up to it, Tracker last-alive dates up to it, and each 2024 core interview from the year in which it was conducted, which is 2025 for many of them. Every point is reconstructed from files released later, so it describes the evidence about events up to the cutoff and not what an analyst could have held at that date. Two checks hold at every cutoff and for each combination of sources. An outcome that is known at one cutoff keeps its value at every later one, and the minimum non-rationalizable share never falls from one cutoff to the next, at any rung, as Proposition 6 requires. The first check holds only after the conflict rule is applied at every date. Without it, 25 respondents, or 0.21 percent of the weight, would be resolved early by a National Death Index death and reopened or reversed once the conflicting source arrives, so a death that conflicts with another source once all information is in never resolves an outcome at any cutoff. With the full record the series reproduces the headline coding exactly, and with the RAND sources alone it reproduces the RAND coding exactly.

Between the 2024 cutoff and the full record no outcome changes value and none is lost. The outcomes that resolve belong to 204 respondents whose 2024 core interview took place in 2025 and 5 whose last Tracker contact is dated 2025, all resolved as having reached the target. Adding the first group alone moves the ladder from the 2024 point to the headline values at every rung, and the second group leaves it unchanged. No National Death Index death is dated after 2024.

The exact test of Section 4 is also recomputed at every cutoff, on the same cells and with the same Holm adjustment. With evidence dated by 2006 it does not reject, since 95.5 percent of the sample is still unresolved. With evidence dated by 2008 it rejects with a smallest adjusted p-value

of 5.7×10^{-4}, and the p-value falls at every later cutoff. With the full record it reproduces the headline value.